

Mergers, Lobbying, and Elections: Is there a "Curse of Bigness"?*

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Abstract

We study the impact of mergers on *quid-pro-quo* lobbying and elections in a political agency model. Two incumbent firms can lobby an incumbent politician to block a pro-competitive reform. The politician's type determines whether they are susceptible to the firms' influence or not. A representative voter tries to infer the politician's type monitoring the policy-making process. We show that lobbying increases when firms merge because rents from political protection are not dissipated by price competition. While greater market concentration may increase prices and political influence, it also improves voters' ability to screen bad politicians by observing distorted policy outcomes. This generates a novel trade-off: mergers can harm consumers through market and political power, yet improve selection of politicians. We characterize when standard consumer welfare-based merger control is too lenient or too strict once these political economy effects are taken into account.

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1 Introduction

Firms' political influence is often regarded as a source of economic inefficiency and democratic failure. Lobbying and regulatory capture can distort policy away from consumers' interests, weaken competition, and entrench incumbent firms. For this reason, firms' influence is typically treated as harmful in both economic analysis and policy debates.

This paper emphasizes that firms' political influence – triggered by changes in the market structure – can also play an informational role. We study a political agency environment in which firms exert influence to obtain favorable policy, politicians differ in their susceptibility to influence, and voters try to infer politicians' types by monitoring the policy-making process. In this setting, lobbying generates a screening mechanism: politicians who accept influence reveal themselves through distorted policies, while politicians who resist influence signal their quality. As a result, voters' selection of politicians improves when policy outcomes are distorted in the short run.

The main contribution of the paper is to show how this political screening channel interacts with changes in the market structure. To this end, we embed a model of political agency into an industrial organization framework in which firms may merge and attempt to block a pro-competitive reform. Firms decide whether to seek political influence based on the rents generated by policy distortions, and politicians decide whether to accept influence based on electoral incentives. This allows us to study how mergers affect both market and political outcomes.

Mergers play a key role in this environment because they alter the scale of the rents associated with regulatory protection. By internalizing competitive spillovers, a merger can increase firms' incentives to seek influence and thus increase voters' ability to select politicians. Importantly, we do not assume that larger or more concentrated firms lobby more. Lobbying arises endogenously in equilibrium and, in some parameter regions, occurs *only* following a merger. This mechanism provides a disciplined sense in which market concentration can translate into political power,

often emphasized in neo-Brandeisian critiques of antitrust. However, our analysis departs from that literature in an important way. Political power is not necessarily welfare-reducing in our framework. While merger-induced lobbying can raise prices and block pro-competitive reforms, it can also improve the selection of politicians.

Our model allows to provide a conceptual discussion of the *consumer welfare standard*, which is the current dominant perspective in antitrust enforcement. Current practice follows the trade-off between efficiencies and market power first pointed out by [Williamson \[1968\]](#): mergers should be approved if and only if cost efficiencies are large enough to compensate for the increase in consumer prices due to higher market power, holding market structure constant. This approach has been criticized for being too narrow by the Neo-Brandeisian antitrust movement ([Wörsdörfer \[2023\]](#)). Following the legacy of Associate Justice of the US Supreme Court Louis Brandeis, critics are concerned about the wider risks that market concentration poses for political freedom and democracy.¹ In this view, antitrust should not only contrast market power but also prevent excessively large corporations from influencing democratic institutions to the detriment of citizens.²

We use the model to discuss optimal merger policy. We show that a merger standard based solely on consumer surplus is optimal only when mergers do not alter the political equilibrium. Otherwise, ignoring political economy effects may lead merger control to be either too permissive or too restrictive, depending on the value of improved selection of politicians.

The main workings of the model are as follows. We consider an industry producing three differentiated goods. Goods 1 and 2 are produced by two incumbent firms with

¹See [here](#) for an overview of the Neo-Brandeisians' critiques to the consumer welfare standard. The debate around the consumer welfare standard goes back at least to [Bork \[1978\]](#), and has been revamped by [Khan \[2016\]](#), [Wu \[2018\]](#), [Shapiro \[2019\]](#), and [Tirole \[2022\]](#), among others.

²Empirical research supports this concern, finding that market concentration increases firms' lobbying expenditures in the US ([Cowgill et al. \[2024\]](#), [Moshary and Slattery \[2024\]](#)). Recent empirical evidence also shows that lobbying expenditures and market concentration have both been increasing over time and across industries. See [here](#), [here](#), and [here](#) for evidence about firms' lobbying activity in the US and the EU. The literature on industrial concentration trends, and its causes and consequences, is an active area of ongoing research. See, e.g., [Grullon et al. \[2019\]](#), [Philippon \[2019\]](#), [De Loecker et al. \[2020\]](#), [Kwon et al. \[2024\]](#).

market power. Good 3 is produced by a competitive fringe.

An incumbent politician decides whether to approve a pro-competitive market reform. If approved, the reform increases competitive pressure by making the fringe more aggressive. It can be interpreted as the removal of a barrier to entry that keeps the price of good 3 artificially high, thereby protecting incumbents' profits. The reform benefits consumers and harms the two incumbent firms.

The incumbents can influence policy through *quid-pro-quo* lobbying ([Grossman and Helpman \[1994\]](#)). Politicians are of two types: Good or Bad. A Good politician is un-corruptible and always implements the reform in consumers' interest. A Bad politician values both lobbying transfers (bribes) and re-election, and thus faces the standard trade-off between behaving well to increase re-election chances and accepting bribes at the risk of losing office.

A representative voter chooses whether to re-elect the politician or not. The objective of the voter is to elect a Good politician. However, they do not observe the politician's type. They try to infer the type by imperfectly monitoring policy-making. Imperfect monitoring creates an incentive for moral hazard. Imperfect information creates a problem of adverse selection of politicians.

We show that the game admits two types of equilibria: Pooling and Separating. In a Pooling equilibrium, all politicians implement the reform, preventing learning and allowing a corruptible politician to be re-elected with positive probability. In a Separating equilibrium, the Bad politician blocks the reform in exchange for bribes, which raises prices but reveals the politician's type and mitigates adverse selection.

Before the lobbying stage, the two incumbents may merge. The merger generates efficiency gains by reducing firms' marginal costs but also increases market power. Importantly, it raises the incumbent firms' benefits from obtaining political protection and thus strengthens their incentives to lobby. In some cases, firms lobby *only* if the merger is approved. Then, the merger can change the nature of the political equilibrium, from Pooling to Separating.

Within this framework, we address our main policy question: which mergers are

welfare improving? Should antitrust authorities account for the political effects of market power?

To answer this question, we introduce an antitrust authority that decides whether to approve the merger before firms lobby. The authority cares about consumers and voters, and its optimal policy depends on the political equilibrium. We identify three cases.

First, suppose the equilibrium is always Pooling, regardless of whether the merger is approved. The reform is always implemented, screening never occurs, and the authority cannot affect politicians' selection. In this case, it approves only sufficiently efficient mergers that lower prices, and the consumer welfare standard is optimal.

Second, suppose instead that the equilibrium is always Separating. As in the previous case, the merger has no political effects, and the voter's payoff does not depend on the merger approval. The voter always successfully screens politicians. However, with some probability, nature draws a Bad politician, and the reform is blocked. Relative to the first case, higher efficiencies are required for a merger to be pro-competitive in expectations. This follows from the strategic complementarity of prices. The two merged firms can exploit market power and increase prices more when the price of their competitor's product is high (i.e., when the pro-competitive reform is not approved). As a result, the consumer welfare standard is too lenient.

Finally, suppose the equilibrium is Separating if and only if the merger is approved. Then the merger increases prices not only through greater market power but also by enhancing political influence and causing the reform to fail. Despite this, voters may prefer the merger because it allows them to identify and remove corrupt politicians. The authority's optimal policy then depends on the value of political screening.

If the value of screening is low, the Separating equilibrium is highly inefficient, as it induces Bad politicians to accept bribes and restrict competition. The authority may then reject even pro-competitive mergers, since blocking the merger sustains a Pooling equilibrium with no lobbying and lower prices. As the value of screening rises, the authority tolerates larger price increases to allow the voter's learning, and

if screening is sufficiently valuable, it may even approve anti-competitive mergers to enable voters to defeat corrupt incumbents.³ Our main welfare result is that ignoring political economy effects in merger control leads to both type I and type II errors.⁴

In the Appendix, we test the robustness of our results. We explore different bargaining settings, we develop a dynamic model with endogenous bribes and office values for politicians, and extend our findings to a broad class of supermodular market games.

We contribute to two main streams of literature. The first is the literature at the intersection of industrial organization and the political economy of regulated markets. [Callander et al. \[2022\]](#) show that politicians have an incentive to protect firms from competition and raise barriers. However, politicians strategically do not protect firms too much; otherwise, firms would stop demanding protection. The same authors show that a liberalization policy can be effective or not depending on the *ex-ante* market structure ([Callander et al. \[2023\]](#)). Recent empirical work by [Akcigit et al. \[2023\]](#) shows that, as firms increase in size, they tend to rely more on non-market strategies, such as political connections, to maintain a dominant position. [Akcigit and Ates \[2023\]](#) document the lack of "creative destruction" in the modern US economy, pointing to regulation and weak antitrust enforcement among the different potential explanations. [Kang \[2016\]](#) structurally estimates a rent-seeking lobbying game, where interest groups lobby politicians to obtain favorable regulations. She finds that lobbying has a small effect on a policy's probability of approval. However, since policies are, on average, very valuable for firms, private returns from lobbying expenditures are substantial.

³A natural question is whether mergers are unique in generating political screening, or whether other, potentially less costly, institutional arrangements could do so. In this regard, we do not claim that mergers are necessary or desirable tools for improving democratic accountability. Other instruments - such as transparency requirements, lobbying regulation, or anti-corruption institutions - could in principle generate similar informational effects. Mergers are distinctive because they already fall within the scope of competition policy and simultaneously affect market outcomes and political incentives.

⁴In the baseline model, the value of screening is exogenous. In the Appendix, we endogenize it in a two-period model in which voters seek to avoid appointing Bad politicians because they always block reforms in the final period. In the absence of out-of-market harms beyond price effects (such as corruption, extraction of public resources, etc.), the consumer welfare standard remains too lenient.

Our two most closely related papers are [Cowgill et al. \[2024\]](#) and [Moshary and Slattery \[2024\]](#), which focus empirically on the impact of mergers on lobbying, finding that mergers increase lobbying activity. Our main contribution is to provide the first (to the best of our knowledge) welfare analysis of mergers that takes into account the political economy effects of market concentration.

The second is the literature about political agency with adverse selection and moral hazard. Seminal papers in this literature are [Fearon \[1999\]](#), [Coate and Morris \[1995\]](#), [Maskin and Tirole \[2004\]](#), and [Banks and Sundaram \[1998\]](#).⁵ The key insight of this literature is that voters and politicians can be seen as principals and agents, respectively. Then, imperfect information generates agency problems. We contribute to this literature by linking electoral accountability with market concentration.

The paper proceeds as follows. In Section 2, we illustrate our model setup. In Section 3, we characterize the equilibrium of the model. Mergers and their welfare properties are considered in Section 4. Finally, Section 5 concludes. Proofs and extensions are relegated to the Appendix.

2 Model

We present a simple model of mergers and lobbying. The model intends to capture market and non-market aspects of competition, and for this reason, it has two main blocks that interact with each other: a political economy block and a competition block. In the first one, two firms lobby a politician who can make a decision that impacts their profitability via a market mechanism, and a representative voter appoints the politician. In the second one, these firms compete against each other in the market, as affected by the politician's decision. Firms may merge or not. Our goal is to perform a welfare assessment of the merger taking into account its political economy effects.

⁵For more recent contributions, see [Wolton \[2019\]](#), [Duggan and Martinelli \[2020\]](#), [Sasso and Morelli \[2021\]](#), and [Blumenthal \[2023\]](#), among many others. For political agency papers related to the topic of money in politics and *quid-pro-quo* lobbying, see [Bils et al. \[2021\]](#) and [Daley and Snowberg \[2011\]](#). See also [Schnakenberg and Turner \[2024\]](#) for a recent survey of this literature.

With this in mind, we now detail every component of the model.

In the main text, we make some assumptions to keep the model as simple as possible to convey our intuition. We relax some of these assumptions in the Appendix. We discuss these modeling choices in detail at the end of this section.

Players We consider a market with three differentiated goods. Let $i \in \{1, 2, 3\}$ index the goods and their producers. Goods 1, 2 are produced by two incumbent firms with market power. These are the firms that can lobby. A competitive fringe sells good 3, and they do not lobby. A representative consumer buys the three goods. At the beginning of the game, the two firms (1, 2) can merge or not. We say that $m = 1$ if the merger occurs, and $m = 0$ otherwise. The merger is exogenous but always profitable for the two firms. In Section 4, we allow a benevolent antitrust authority to decide upon the merger approval.

There is an incumbent politician (whom we often refer to as "P"). P can implement a pro-competitive reform that decreases the price of the fringe good p_3 , for instance, by making entry into the market by the competitive fringe easier. The incumbent firms can lobby P trying to avoid that. P has a type $\theta_P \in \{\text{Good}, \text{Bad}\}$. If P is Good, they are not interested in lobbying transfers, such as bribes. If P is Bad, they care about bribes. A representative voter chooses whether to re-elect P or replace them with a challenger. The challenger's type is denoted by $\theta_C \in \{\text{Good}, \text{Bad}\}$.

Policy The price of the fringe good $p_3(a)$ depends on a pro-competitive reform a . We say that the reform is implemented by P if $a = 1$ and it is not implemented if $a = 0$. We assume that $p_3(1) < p_3(0)$. Without loss of generality, let $p_3(1) = 0$ and $p_3(0) = \eta$, with $\eta > 0$.

There are various interpretations for a . For instance, firms 1, 2 may be two domestic firms competing against a fringe of foreign goods whose prices depend on the level of import tariffs. Alternatively, firms 1, 2 may describe two private firms (e.g., hospitals, schools, or pharmacies) competing with a fringe of public providers whose number

and/or quality is affected by political decisions.⁶ As a further example, consider the case of entry regulation. Politicians can restrict entry in many markets (e.g., retail, or ride-sharing), thereby (indirectly) controlling prices. All these examples share a crucial feature of our parsimonious model. Politicians can take actions that influence the degree of competition firms face in the market and, therefore, their profits. Firms, in turn, can lobby politicians to obtain favorable policy.

We discuss the details of the lobbying and merger processes in the following paragraphs.

Timing and Actions The timing of the game runs as follows.

Stage 1 Nature draws θ_P, θ_C .

Stage 2 P chooses whether to implement the reform ($a = 1$) or not ($a = 0$) by committing to a mechanism $a(l_1, l_2)$.

Stage 3 Firms observe the mechanism and choose whether to lobby ($l_i = 1$) or not ($l_i = 0$). The reform a is implemented.

Stage 4 Firms observe $p_3(a)$ and simultaneously set prices p_1, p_2 . The consumer observes prices and chooses a consumption plan d_1, d_2, d_3 .

Stage 5 The voter chooses whether to re-elect P ($r = 1$) or not ($r = 0$). Payoffs are realized.

Payoffs The representative consumer has a quadratic utility function *à la* [Singh and Vives \[1984\]](#):

$$u(d_1, d_2, d_3) = d_1 + d_2 + d_3 - \frac{1}{2} (d_1^2 + d_2^2 + d_3^2) - \gamma (d_1 d_2 + d_2 d_3 + d_1 d_3) . \quad (2.1)$$

⁶This example may be particularly relevant given the recent wave of mergers between US hospitals. See [Gowrisankaran et al. \[2015\]](#) and [Prager and Schmitt \[2021\]](#), among others.

The parameter $\gamma \in (0, 1)$ captures the degree of product differentiation.⁷ If $\gamma = 0$, the three products are independent, and the two incumbent firms act as separate monopolists. If $\gamma = 1$, goods are perfect substitutes, and Bertrand competition brings profits down to zero. In either limiting case, firms would never lobby as the reform a does not impact their profits. Hence, we study the case when $0 < \gamma < 1$, that is, products are imperfect substitutes (Amir et al. [2017]).

The voter cares about electing a Good politician. Let $\theta^*(r)$ be the type of politician winning the election. Then,

$$\theta^*(r) = \begin{cases} \theta_P & \text{if } r = 1 \\ \theta_C & \text{if } r = 0. \end{cases} \quad (2.2)$$

At the end of Stage 5, $\theta^*(r)$ is revealed to the voter. The voter's payoff is:

$$v(r) = \begin{cases} -\phi & \text{if } \theta^*(r) = \text{Bad} \\ 0 & \text{if } \theta^*(r) = \text{Good} , \end{cases} \quad (2.3)$$

where $\phi > 0$ measures the benefits associated with screening. A possible interpretation for ϕ is as follows. Suppose that a Bad politician is re-elected. While in office, a corruptible politician may adopt policies that appeal to specific interest groups (perhaps in exchange for bribes) rather than voters. In this context, ϕ represents the cost of such actions to the voters in present-value terms.

The profits of firms $i \in \{1, 2\}$ are:

$$\pi_i(p_i, d_i, l_i) = d_i [p_i - c(m\mu + (1 - m))] - tl_i , \quad (2.4)$$

where d_i is the quantity sold by firm i , $c > 0$ is the (symmetric) marginal cost, $\mu \leq 1$ captures the efficiencies activated by the merger, and $t > 0$ is the value of the

⁷Assume that the consumer has a quasi-linear utility function, where (2.1) is the non-linear part. Then, the demand for goods 1, 2, 3 does not depend on income, and a partial equilibrium analysis is justified (Singh and Vives [1984], Motta [2004], Choné and Linnemer [2020]).

bribe. We denote the firm's *operating profits*, profits net of lobbying expenditures, by:
 $\hat{\pi}_i(p_i, d_i, l_i) = \pi_i(p_i, d_i, l_i) + t l_i$.

We assume that the merger can generate merger-specific efficiencies from the joint production of goods 1, 2. We model this by assuming that if the merger goes ahead ($m = 1$), marginal costs are μc , where $\mu \leq 1$. If $m = 1$, the merged entity chooses prices p_1, p_2 and lobbying efforts l_1, l_2 to maximize the joint sum of profits from selling the two goods. Notice that this framework allows for a meaningful welfare assessment, as the merger entails a trade-off: market power always increases, but there may be a countervailing force if efficiencies are large enough (Williamson [1968]).

The Good politician P is a behavioral type. They always implement the reform: $a = 1$. One interpretation is that they are ethical agents who care about decreasing prices in the interests of consumers. The Bad P's payoff is:

$$U_{\text{Bad}}(r, l_1, l_2) = rV + (l_1 + l_2)t, \quad (2.5)$$

where $V > 0$ captures the value of the office, if re-elected, and $t(l_1 + l_2)$ is the total amount of bribes received from firms. In our model, *quid-pro-quo* lobbying takes the form of transfers. In reality, there are various possible interpretations for t : revolving doors, bribes, donations to parties, informational benefits, etc. In the remainder, we simply refer to bribes as a shorthand for all these alternatives but we do not take a position on the legality of lobbying or bribery.⁸

Information Structure The voter does not observe politicians' types θ_P, θ_C . Let $q \in (0, 1]$ be the voter's prior belief that $j = \text{Good}$, where $j \in \{\theta_P, \theta_C\}$. We assume that m is common knowledge: everyone knows whether the merger occurred or not.

With probability $x \in (0, 1]$, the voter observes the action a . There are two cases. The first case is $x = 1$. In this case, the voter and the consumer can be thought of as the same player. This player's payoff would be given by the sum of (2.1) and (2.3). In

⁸See Bombardini and Trebbi [2025] for a review of the various channels of political influence of large corporations.

fact, if $x = 1$, the voter and the consumer have the same information.

The second case arises when $x < 1$. In this case, there is imperfect monitoring and, therefore, moral hazard (Blumenthal [2023]). The voter and the consumer are two different players. There is at least one possible reduced-form interpretation. The parameter x may capture the transparency of the political system and/or the quality of the media. When x is high, policy-making is transparent as media inform citizens about the decisions taken by politicians. When x is low, instead, citizens are not well informed about politics. In this context, the voter may possess different information than the consumer (the consumer can always infer a from observing p_3). For instance, the voter may not be directly interested in the consumption of goods 1, 2, 3, but still be interested in learning θ_P . Moreover, even if the voter is also a consumer, prices could be not perfectly informative.⁹ The extent of moral hazard, measured by $1 - x$, determines how likely P can accept bribes and still get re-elected. Only in the limiting case when $x \rightarrow 0$, the voter does never observe P 's action, and the Bad politician does not trade off re-election and bribes.

Discussion of the Main Assumptions We now discuss the main assumptions that we made in order to keep the model as simple as possible.

Exogenous lobbying transfers and politicians' power of commitment. In our model, we assume that lobbying transfers are exogenous, and that firms choose only whether to lobby or not. When firms reach an agreement with P , the transfer t is transferred one-for-one. First, note that if this transfer provides a benefit of value t for P , the cost to firms may actually be αt , where $\alpha \leq 1$. For simplicity and without loss of generality, we fix $\alpha = 1$. Moreover, we let P have the power to commit to a TIOLI offer for the two firms. Neither of these two assumptions is consequential for our results.

⁹Indeed, in many real-world examples, prices can be "high" not because of a lack of competition but because of macroeconomic shocks or external factors. Therefore, if consumers (voters) observe high prices, they may be uncertain about the causes behind such prices. For instance, during the 2022-2023 inflation wave, there was an intense debate on the causes of such inflation, with some experts attributing them mainly to "external factors," such as the War in Ukraine or the Covid 19 Pandemic; and others pointing to companies' opportunistic behaviors ("greedflation"). This imperfect information makes prices not (fully) informative and hence generates imperfect monitoring from the point of view of consumers and a problem of moral hazard from the politicians' perspectives.

In the Appendix, we generalize the model to different bargaining structures. In particular, in Appendix C.2, there are endogenous lobbying transfers; while in Appendix C.1, the two firms have the power of committing to a TIOLI offer to P. The key assumption is that agreements cannot be re-negotiated. For instance, P cannot accept bribes promising to block the reform, and then fail to keep their promise. This is in line with many political economy models, where politicians commit to policies before elections (think, for instance, to the tradition of Downsian models of elections).

Distinction between voter and consumer under exogenous value of screening. We distinguish between a representative voter and a representative consumer. This captures imperfect monitoring ($x < 1$), whereby market outcomes are not always fully reflected in voting decisions. While it would be natural to identify consumers with voters, this distinction allows us to model limited accountability without complicating the analysis, and to highlight the trade-off between worse market outcomes (higher prices due to lobbying) and improved selection of politicians. Consumers care only about prices, whereas voters care about selecting the right politician.

In Appendix C.2, we study a dynamic version of the model with endogenous bribes, in which the voter and the consumer are the same player (to this end, we assume $x = 1$). In this version of the model, the voter (who is also a consumer) cares about the selection of politicians because Bad politicians always accept bribes and block pro-competitive reforms during the last period of office. This logic is standard in political agency models. The key assumption is that voters care about the type of elected politicians, and have a preference for electing un-corruptible politicians.

Demand function and market environment. For tractability, we adopt a specific consumer utility function and model lobbying as an effort to keep a rival's price artificially high. In Appendix B, we generalize our results to a broad class of supermodular games. We show that mergers increase incentives to lobby because of the supermodular nature of Bertrand competition. The key assumption is that lobbying targets a common component: both firms benefit when the price of the fringe good is high.

Other assumptions. Our results also rely on a few additional assumptions worth

discussing. First, we model lobbying as a *quid-pro-quo* exchange, whereby politicians grant favors in return for private benefits, such as monetary bribes, non-monetary perks, or post-term employment. We do not consider informational lobbying, in which interactions with firms provide policymakers with valuable information.¹⁰

Second, in our model lobbying transfers always reduce the incumbent’s re-election prospects. With positive probability $x > 0$, voters observe the policy-making process and infer that an incumbent who blocks the pro-competitive reform is of the Bad type. This implies that an increase in lobbying following a merger can improve political screening. By contrast, if lobbying transfers could be used to enhance re-election (e.g., if voters never observed policy choices, $x = 0$, and politicians used bribes to finance advertising) mergers would weaken voters’ ability to infer politicians’ types.

Third, we adopt a partial equilibrium approach. While voting and market competition are general-equilibrium phenomena, a stylized single-industry framework can still be informative about the incentives of politicians, voters, and firms in environments where large corporations use political influence to obtain favors. Telecommunications provides a natural example: the industry has experienced repeated mergers among large operators and is characterized by extensive lobbying and political connections. Consistent with our modeling assumptions, [Faccio and Zingales \[2022\]](#) show that politically connected incumbents face weaker pro-competitive regulation and charge higher prices.

3 Results

Our solution concept is Perfect Bayesian Equilibrium (equilibrium henceforth) in pure strategies. A strategies–belief pair is an equilibrium if and only if:

1. Each player’s strategy maximizes their expected payoff given all the other players’ strategies and beliefs;

¹⁰For models of informational lobbying, see [Schnakenberg \[2017\]](#), [Awad and Minaudier \[2025\]](#), and [Besanko and Spulber \[1993\]](#) for an application to antitrust. [Bombardini and Trebbi \[2020\]](#) review the empirical literature, distinguishing between *quid-pro-quo* and informational lobbying.

2. For any observation of P's action, the voter's belief is updated via Bayes' Rule, and their action is optimal given the equilibrium beliefs.

We follow the convention that starred variables denote equilibrium actions.

In this section, we take the merger m to be exogenous, and we study the impact of the merger on prices, lobbying, and elections. In Section 4, we study how these effects interplay with each other in the welfare analysis of the merger.

3.1 Preliminary Results

Let us start from the election stage (Stage 5). Let $\hat{q}$ denote the voter's posterior belief that $\theta_P = \text{Good}$. Let us assume that, in case of indifference, the voter chooses to re-elect the incumbent.¹¹ The voter's optimal re-election rule is:

$$r^* = \begin{cases} 1 & \text{if } \hat{q} \geq q \\ 0 & \text{otherwise} . \end{cases} \quad (3.1)$$

If the Bad P chooses $a = 0$ and the voter observes it, then $\hat{q} = 0$. Therefore, they are re-elected with probability $1 - x$. If they choose $a = 1$, they are re-elected with probability 1. P then faces a trade-off between behaving well today and ensuring re-election tomorrow, or accepting bribes and giving up the office (with some probability).

Let us now consider the market stage. In Stage 4, firms set prices anticipating the consumer's demand functions. The representative consumer's demands for the three goods $i, j, k \in \{1, 2, 3\}, i \neq j \neq k$ are:

$$d_i = \frac{1 - \gamma - p_i(\gamma + 1) + \gamma(p_j + p_k)}{(1 - \gamma)(2\gamma + 1)} . \quad (3.2)$$

¹¹ Our findings remain qualitatively unchanged under any change of the tie-breaking rule where there is a strictly positive probability that the incumbent is re-elected. If this were not the case (i.e., if the voter always elected the challenger when indifferent), the Bad incumbent would face no trade-off between bribes and re-election. In particular, if the Bad incumbent could never be re-elected, they would always accept bribes. Notice that this situation would arise if, from the voter's prior perspective, the challenger were more likely to be of a Good type than the incumbent. For this reason, we assume that the incumbent and the challenger are *ex ante* identical and that, whenever indifferent, the voter re-elects P with some positive probability

We obtain (3.2) by simply maximizing (2.1) given prices. As $\gamma > 0$, the demand for good i increases in the price of goods j, k .

In Appendix A.1, we obtain subgame equilibrium prices as a function of p_3 and market structure m (i.e., with and without the merger m). We omit these computations from the main text. The important results from Appendix A.1 are as follows. First, for any market structure m , equilibrium prices increase in p_3 (and thus decrease in a). This result stems from prices' strategic complementarity. Second, equilibrium profits of firms 1, 2 decrease in a , as they face a stronger fringe. Finally, we compare prices with and without the merger. Unsurprisingly, the merger decreases prices if and only if efficiencies are high enough, that is, if μ is low enough. If the reform is not implemented ($a = 0$), higher efficiency (lower μ) is required for the merger to be pro-competitive. This result will play a key role in the merger welfare assessment. When the merger induces P to adopt the non-competitive policy ($a = 0$), a higher level of efficiency must be achieved by the merger to be pro-competitive.

3.2 Lobbying Equilibrium

We now consider the lobbying game (Stage 2 and Stage 3). For all m , there are two possible types of equilibria: Pooling and Separating. In a Pooling equilibrium, both types of P implement the reform. The consumer enjoys lower prices, but the voter does not learn P's type and, with some probability, re-elects the Bad P. In a Separating equilibrium, the Bad P does not implement the reform and both firms lobby. With probability x , the voter observes the action, and learns P's type. We first characterize a Separating equilibrium, and then we discuss conditions under which such an equilibrium exists.

In a Separating equilibrium, the Bad P commits to the mechanism $a^*(1, 1) = 0$ and both firms choose $l_i^* = 1$. Suppose that P does not offer protection, then no firm wants to lobby. In the same way, if neither firm lobbies, P ensures re-election by pooling. Then, any (possibly) optimal contract implies protection in exchange

for some lobbying. We are left to consider the alternative mechanism $a^*(1, 0) = 0$. We show that this cannot be optimal for P. In fact, P knows that if $l_i = 1$ is incentive compatible (IC) for firm i , $l_j = 1$ is also IC for firm j . Therefore, the contract $a^*(1, 0) = 0$ is strictly dominated. If P can "sell" protection ($a = 0$) for $2t$, there is no reason to give out a 50% discount and accept a price of t . This result depends on the fact that P has the power of commitment to the offer. In Appendix C, we show that if firms can commit to the offer, there is an equilibrium where only one firm lobbies.¹²

When is a Separating equilibrium possible? Let $p^m(a)$ denote the vector of equilibrium prices as a function of a and m . For $a = 0$ to be individually rational (IR) for the Bad P, it must be that:

$$(1 - x) V + 2t \geq V . \quad (3.3)$$

P knows they can ensure re-election by pooling ($a = 1$), which yields a payoff of V . By separating, P gets bribes ($2t$) but wins the office only with probability $1 - x$. P trades off tomorrow's re-election chances with today's bribes.

Paying bribes must be incentive-compatible (IC) for individual firms.¹³ Let

$$\Delta\pi(m) = \begin{cases} \hat{\pi}_i(p^{m=0}(0)) - \hat{\pi}_i(p^{m=0}(1)) & \text{if } m = 0 \\ \frac{1}{2} [\sum_i \hat{\pi}_i(p^{m=1}(0)) - \sum_i \hat{\pi}_i(p^{m=1}(1))] & \text{if } m = 1 \end{cases} \quad (3.4)$$

be the return for firms from blocking the pro-competitive reform, as a function of market structure m . We derive the expressions of $\Delta\pi(m)$ in Appendix A.2. Firms' IC requires:

$$t \leq \Delta\pi(m) . \quad (3.5)$$

¹²The mechanism $a^*(1, 1) = 0$ also solves the coordination problem for firms as they both know they need to lobby to induce $a = 0$. When the politician is willing to accept a single bribe, and firms offer the contract, there is a coordination problem arising from the multiplicity of equilibria. In this case, the merger increases lobbying also because it solves such coordination problem. Our main conclusions are robust to this extension, which we study explicitly in Appendix C.1.

¹³For simplicity, we rule out the case in which the Bad politician accepts a single bribe of value t that the two firms coordinate to split. While this case may be realistic under a merger, we exclude it in order to facilitate comparison with the no merger case. Our qualitative results, however, are not affected by this.

Combining P's IR and firms' IC, we obtain the following result.

Proposition 1. *In a Pooling equilibrium, both P's types implement the pro-competitive reform ($a^* = 1$). In a Separating equilibrium, a Bad P does not implement the reform ($a^* = 0$), and both firms lobby ($l_1^* = l_2^* = 1$). Moreover,*

(i) *For all $m \in \{0, 1\}$, the equilibrium is Separating if and only if*

$$\frac{xV}{2} \leq t \leq \Delta\pi(m) .$$

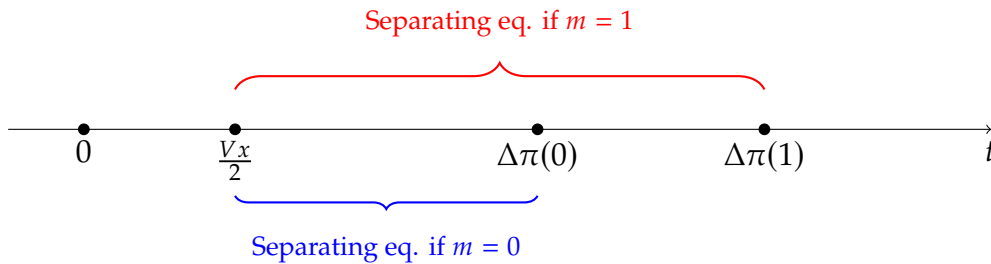
In a Separating equilibrium, the voter learns P's type with probability x . Otherwise, the equilibrium is Pooling, and the voter does never learn P's type.

(ii) *The merger increases firms' incentives to lobby:*

$$\Delta\pi(1) \geq \Delta\pi(0) .$$

Figure 3.1 illustrates the findings of Proposition 1.

Figure 3.1: Existence of Separating Equilibrium



First, a Separating equilibrium exists if and only if the cost of lobbying t is neither too high nor too low.¹⁴ If t is too low, bribes are not sufficiently valuable, and the Bad P prefers to ensure re-election by pooling. If t is too high, lobbying is not worth it for firms.

¹⁴In Appendix C, we show the conditions for the existence of a Separating equilibrium when t is endogenous.

Second, a merger increases firms' incentives to lobby as it increases the returns from protection. The intuition is simple. Lobbying to fend off competition from the fringe is valuable, as it can increase rents to firms 1, 2. Without a merger, however, these rents are partly dissipated by the rivalry between 1, 2. With a merger, this dissipation is muted, and therefore merging firms are willing to pay more for protection. In Appendix B, we generalize this intuition to a more general setting.¹⁵

The increase in lobbying after a merger is not driven by the public good nature of lobbying, as in Moshary and Slattery [2024], or by economies of scale. In our model, by muting price competition, the merger creates additional rents that trigger firms' incentives to lobby. As a consequence, even if firms could coordinate to maximize the total sum of profits in the lobbying game, as would happen if firms exerted influence via industry-level associations, our result would still go through.¹⁶ In other words, the merger increases firms' incentives to lobby because prices (and not lobbying efforts) are set non-cooperatively.

We now discuss the comparative statics of the upper and lower bounds in Figure 3.1. The lower bound $\frac{Vx}{2}$ does not depend on market structure. The merger does not impact the politician's incentives to accept the bribes. This lower bound increases as x and V increase. When x is low, moral hazard bites: the Bad P knows that by separating, they can enjoy bribes and still be re-elected with a relatively high probability. Therefore, the interval for t that supports a Separating equilibrium decreases in x . On the contrary, as V increases, holding the office is more valuable, and the Bad P has a higher incentive to ensure re-election by pooling.

From the firm's perspective, the merger increases the incentives to lobby. The upper bound $\Delta\pi(1)$ decreases in μ . The more efficient the merger, the higher the incentives to lobby, as the the returns internalized from protection are higher. In the same way, for all m , $\Delta\pi(m)$ is also increasing in η . The higher the threat from the fringe, the higher the incentives to lobby.

¹⁵In particular, we show that Proposition 1 extends to any supermodular game with spillovers.

¹⁶To see this, simply note that blocking the pro-competitive reform increases industry profits more after a merger: $2\Delta\pi(0) < 2\Delta\pi(1)$.

Equilibrium Uniqueness The equilibrium described in Proposition 1 is unique among the class of equilibria in pure strategies. Uniqueness is guaranteed by our assumption on the tie-breaking rule for the voter in Stage 5 and by the following refinement. Whenever there are multiple equilibria in the lobbying game (Stage 2-Stage 3), we select the voter-preferred equilibrium.

The only potential multiplicity of equilibria arises when t equals either the lower or the upper bounds of the interval supporting separation in Proposition 1. For a given m , inside the interval, the Separating equilibrium is unique. Outside the interval, Separating is either not IC for firms or not IR for P. Hence, the unique equilibrium is Pooling. At the boundaries of the interval, either the two firms or the politician are indifferent between their actions. Given our refinement, we always select the Separating equilibrium over a possible Pooling one.¹⁷

4 Merger Assessment

We now provide a welfare assessment of the merger. That is, we compare players' welfare if $m = 1$ and $m = 0$. Our model captures the market and non-market effects of the merger, via prices and lobbying respectively. As shown above, a merger increases firms' lobbying activity and their ability to influence policy-making. On the one hand, this political power is bad for the consumer as it further increases prices on top of the usual market power effect. On the other hand, the additional political power is good for the voter when it allows them to learn about P's type. We now discuss how this new trade-off interplays with the traditional price effects in the merger assessment.

To this end, we introduce an additional player: a benevolent antitrust authority. At the beginning of the game (before Stage 1), the competition authority decides whether to approve the merger ($m = 1$) or not ($m = 0$) by maximizing the sum of consumers' and voters' payoffs. Let us recall that if $x = 1$, we can think of the voter and the

¹⁷In this way, we also rule out possible equilibria in mixed strategies. In the market game (Stage 4), the equilibrium is unique by the strict concavity of payoffs.

consumer as a unique player. If $x < 1$, the two players are distinct as they possess different information.¹⁸ This objective function of the antitrust authority is based on the impact on citizens (consumers and voters) and does not take into account firms or the politician. In this regard, our authority’s mandate goes beyond what competition authorities usually do. Their remit is typically to maximize the welfare of the consumers in the market affected by the transaction, which goes under the name of *consumer welfare standard*, as we anticipated in the Introduction. It is important to remark that we are not claiming that competition authorities *do* or *should* start to consider the direct impact of mergers on elections and lobbying. Our results only illustrate the broader welfare implications of merger policy and allow us to contribute to the debate on what antitrust authority *could* do. At the end of this Section, we provide a possible way to interpret our findings in practice.¹⁹

Notice that we also assume that firms cannot lobby the antitrust authority. Specifically, firms 1, 2 cannot lobby to get the merger approved. This simplifying assumption is based on the empirical findings of Cowgill et al. [2024]. They find that merging firms in the US do spend considerable lobbying money but mostly in the Congress. They also do not find anticipatory effects before the approval date of a merger, suggesting that the main targets of lobbying money are not antitrust enforcers. Moreover, the objective of our model is to analyze how market concentration shapes firms’ ability to influence politics and how this affects optimal merger control. To this end, the benevolent competition authority is introduced as a rhetorical device to conduct a welfare analysis rather than an actual player of the game. We do not model the political pressure faced by competition authorities in reality, as this is beyond the scope of our paper. However, this is a natural avenue for further research on the broad topic

¹⁸In a more realistic setting, there are multiple consumers and voters. If the sets of consumers and voters are disjoint or perfectly overlapped, a welfare function that sums voters’ and consumers’ payoffs implies no bias due to the double-counting of players. However, if only some consumers were also voters, this welfare measure would double-count the overlapping players. However, our results are qualitatively robust to players’ re-weighting in the welfare function.

¹⁹See Motta [2004] for a discussion about antitrust authorities’ mandates and, in particular, for an introduction to the debate around the consumer welfare standard. Currently, in both the US and in the EU, competition authorities follow, albeit with some differences, a consumer welfare standard.

of the political economy of competition policy.²⁰

We start by writing down the authority's payoff across the different equilibria. We suppose that the authority does not observe θ_P . Thus, they do not know what type of P will be at play, but they know what kind of equilibrium will occur and, therefore, how different types of P would behave. We assume that the authority shares the prior beliefs of the voter about P's type. Let $W(m)$ denote the authority's payoff as a function of market structure. Let also $d^*(p^m(p_3))$ be the vector of demands. In a Pooling equilibrium,

$$W_{Pool}(m) = u(d^*(p^m(0))) + (1 - q)(-\phi) . \quad (4.1)$$

In a Pooling equilibrium, the consumer always enjoys lower prices, but the voter always re-elects P. Then, with probability $1 - q$, the voter re-elects a Bad P and pays the penalty $-\phi$.

In a Separating equilibrium,

$$W_{Sep}(m) = q[u(d^*(p^m(0)))] + (1 - q)[u(d^*(p^m(\eta))) + (1 - x)(-\phi) + x(1 - q)(-\phi)] . \quad (4.2)$$

With probability q , P is Good, and the consumer enjoys lower prices. It does not matter if the voter observes a or not; in either case, the voter re-elects the incumbent and pays no penalty. With probability $1 - q$, P is Bad, and the reform is not implemented. In this case, with probability x , the voter observes $a^* = 0$, and the Bad P is defeated. The newly elected politician is Bad with probability $1 - q$. With probability $1 - x$, the action is not observed, and the Bad P is re-elected. We show the expressions of (4.1) and (4.2) in Appendix A.3.

Lobbying comes at the cost of high prices today, but it increases the probability that Bad politicians are defeated tomorrow. In particular, in a Pooling equilibrium,

²⁰See Motta and Ruta [2012] for a political economy model of mergers affecting international markets, in which firms can lobby and governments can influence antitrust authorities.

the voter appoints a Bad politician with probability $1 - q$. In a Separating equilibrium, a Bad P is appointed with probability

$$(1 - q) [(1 - x) + x (1 - q)] < 1 - q . \quad (4.3)$$

Therefore, the voter always prefers a Separating equilibrium over a Pooling equilibrium. The opposite is true for the consumer. The authority needs to trade off these two effects as well as to anticipate the effect of their decision on the likelihood that lobbying will occur in equilibrium.

Let us define two thresholds for μ , corresponding to the different levels of efficiencies required for prices to decrease after a merger, and as a function of the policy a :

$$\begin{aligned} \bar{\mu}(0) : \mu \leq \bar{\mu}(0) &\Rightarrow p^{m=1}(0) < p^{m=0}(0) \\ \bar{\mu}(\eta) : \mu \leq \bar{\mu}(\eta) &\Rightarrow p^{m=1}(\eta) < p^{m=0}(\eta) . \end{aligned} \quad (4.4)$$

When the pro-competitive reform is not approved, the merger must be more efficient to decrease prices, that is, $\bar{\mu}(\eta) < \bar{\mu}(0)$.²¹ This result plays a role in the characterization of the optimal merger policy. The intuition is that, because of strategic complementarity of prices, the merged firm can exploit market power more (and thus further increase prices) when the fringe good is not very appealing for the consumer (because of its high price).²² When, instead, the price of the fringe good is low because the pro-competitive reform is implemented, additional competitive pressure from the fringe limits the merged firm's ability to raise prices.

A traditional competition authority, assuming that politicians are not corrupt and considering only price effects, would approve the merger if and only if $\mu \leq \bar{\mu}(0)$. We

²¹See Appendix A.1 for the expressions of (4.4).

²²In Appendix B, we show that this follows from the supermodularity of the Bertrand game. See Footnote 38.

refer to this approach as the consumer welfare standard.²³

To describe the decision of the authority and how that compares with the consumer welfare standard, we distinguish three cases corresponding to the different regions in Figure 3.1.

(a) $t \in [0, \frac{V_x}{2})$ or $t > \Delta\pi(1)$. For any m , the equilibrium is always Pooling. Political effects do not matter: there is no lobbying, and the Bad P is always re-elected with probability $1 - q$. All P types approve the reform: $a^* = 1$. Thus, the authority approves the merger if and only if $\mu \leq \bar{\mu}(0)$. In this case, the consumer welfare standard is optimal.

(b) $t \in [\frac{V_x}{2}, \Delta\pi(0)]$. For any m , the equilibrium is always Separating. As in the previous case, the probability of election of a Bad politician does not depend on m . The authority approves the merger if and only if μ is low enough for prices to decrease. However, with probability $1 - q$, a Bad P is in office, and the reform is not implemented ($a^* = 0$). The optimal standard to be adopted would be $\bar{\mu}(\eta)$. Therefore, the authority's optimal merger policy is stricter than a consumer welfare standard that ignores lobbying that does happen. The critical threshold, $\hat{\mu}$, is such that $\hat{\mu} \in [\bar{\mu}(\eta), \bar{\mu}(0)]$.²⁴ Clearly, $\hat{\mu} \rightarrow \bar{\mu}(\eta)$ if $q \rightarrow 0$ and $\hat{\mu} \rightarrow \bar{\mu}(0)$ if $q \rightarrow 1$.

(c) $t \in (\Delta\pi(0), \Delta\pi(1)]$. In this case, there is a Separating equilibrium if the merger is approved ($m = 1$) and a Pooling equilibrium if the merger is not approved ($m = 0$). If $m = 0$, the consumer enjoys lower prices (the reform is always

²³Competition authorities typically take merger decisions without accounting for dynamic effects on market structure, such as merger waves. For instance, when two mergers occur consecutively in the same market but are notified at different dates, the European Commission applies the priority rule, vetting the first transaction as if the second will not occur, effectively freezing market structure at the first notification date. Similarly, in our model, a traditional consumer welfare-focused authority ignores dynamic effects (the block of pro-competitive reforms arising from the political power firms gain through mergers). See [Nocke and Whinston \[2010\]](#) for a dynamic model of merger review and a discussion of the limits of a static consumer welfare approach.

²⁴To see this, suppose $\hat{\mu} > \bar{\mu}(0)$. Then, if $\mu \in (\bar{\mu}(0), \hat{\mu})$, the merger increases prices with probability 1 but it is accepted, which cannot be optimal for the authority. Analogously, say $\hat{\mu} < \bar{\mu}(0)$. Then, if $\mu \in (\hat{\mu}, \bar{\mu}(0))$, the merger decreases prices with probability 1, but it is rejected, which cannot be optimal for the authority.

implemented), but the voter does not learn about P's type. If $m = 1$, with probability $1 - q$, the consumer faces higher prices in the market, but the voter learns P's type (with probability x), thereby defeating the corrupt politician. The authority faces a trade-off between having the Good reform today and being able to screen P types tomorrow. This new trade-off augments the problem of the authority, otherwise only trying to balance market power and efficiencies. The authority approves the merger if and only if $\mu \leq \tilde{\mu}$, but potentially $\tilde{\mu} < \bar{\mu}(0)$ or $\tilde{\mu} > \bar{\mu}(0)$. In particular, $\tilde{\mu} > \bar{\mu}(0)$ when ϕ is high enough; and $\tilde{\mu} < \bar{\mu}(0)$ when ϕ is low enough.²⁵ In this case, the consumer welfare standard can be either too strict or too weak, depending on the benefits of separation ϕ .

When ϕ is low, the merger must be very efficient ($\tilde{\mu} < \bar{\mu}(0)$) to be approved. The intuition is simple. With probability $1 - q$, P is Bad, and the merger gives firms enough power to influence P's behavior. As a result, if the merger is approved, prices increase (in expectations) because of firms' political power rather than market power. The voter's benefits from separating P types do not offset the loss in consumer utility. It can also be the case that $\tilde{\mu} < \bar{\mu}(\eta)$.²⁶ Efficiencies must be high enough not only to decrease prices, but also to compensate for the fact that, if the merger were not allowed, the consumer would always enjoy the pro-competitive reform, no matter the type of P.

As ϕ increases, welfare in the Separating equilibrium increases. The authority starts to trade off more of today consumer surplus to allow the voter to defeat the corrupt politician tomorrow. When ϕ is high enough, the authority can even allow mergers that increase prices ($\tilde{\mu} > \bar{\mu}(0)$), as the increase in the voter's payoff fully compensates for the decrease in the consumer's utility.²⁷

The following Proposition summarizes the results of this Section.

²⁵We show this formally in Appendix A.3 and A.4.

²⁶See Appendix A.3 and A.4.

²⁷At the end of this Section and in Appendix C.2, we discuss conditions under which this can be the case in practice.

Proposition 2. *The authority allows the merger ($m^* = 1$) if and only if $\mu \leq \mu^*$, where*

$$\mu^* = \begin{cases} \bar{\mu}(0) & \text{if } t \in [0, \frac{V_x}{2}) \text{ or } t > \Delta\pi(1) \\ \hat{\mu} & \text{if } t \in [\frac{V_x}{2}, \Delta\pi(0)] \\ \tilde{\mu} & \text{if } t \in (\Delta\pi(0), \Delta\pi(1)] , \end{cases}$$

$\hat{\mu} \in [\bar{\mu}(\eta), \bar{\mu}(0)]$, and $\tilde{\mu} \gtrless \bar{\mu}(0)$ depending on ϕ .

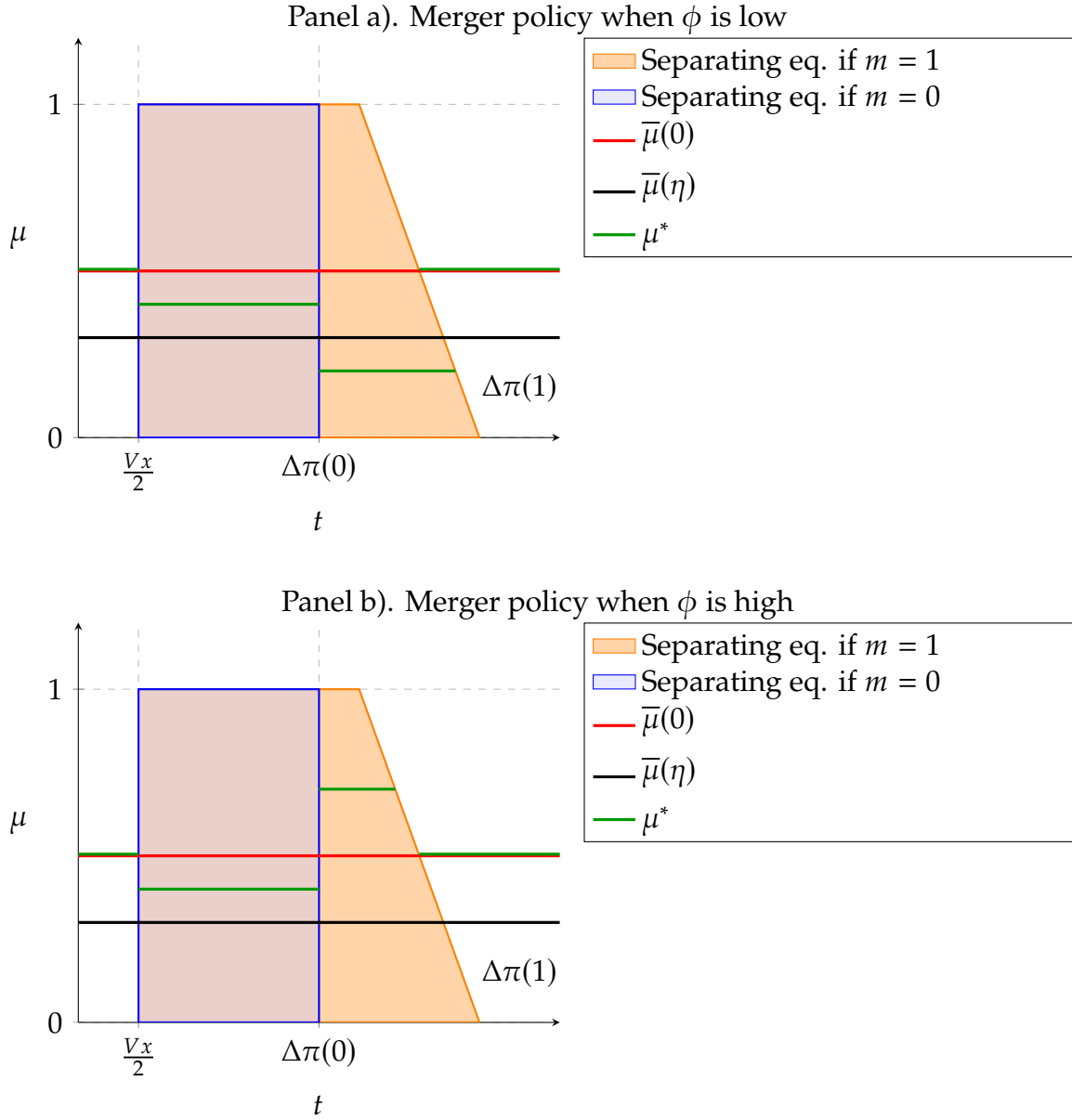
Proposition 2 captures the key insights of our model.²⁸ We summarize the results with the help of Figure 4.1, which draws the efficiency thresholds in various regions as a function of the level of bribes and ϕ . Figure 4.1-panel a) corresponds to the case of low ϕ . Figure 4.1-panel b) corresponds to the case of high ϕ .

The optimal merger policy coincides with the consumer welfare standard when the equilibrium does not exhibit lobbying, that is, when the equilibrium is Pooling for all m (white region).

The optimal merger policy is stricter than the consumer welfare standard when the equilibrium exhibits lobbying, but the merger does not induce the voter's learning, that is, when the equilibrium is Separating for all m (blue region). In the same way, the optimal policy is stricter than the consumer welfare standard when the merger induces a Separating equilibrium, but the benefits from the voter's learning do not offset the increase in prices (orange region in Figure 4.1-panel a)). When, instead, the merger can induce learning, and benefits from the voter's learning are high enough (orange region in Figure 4.1-panel b)), the optimal merger policy can be more lenient than what the consumer welfare standard would suggest.

²⁸The expressions for the different thresholds in Proposition 2 are reported in Appendix A.3.

Figure 4.1: Optimal merger policy



Discussion We have shown that the traditional consumer welfare standard can be sub-optimal when political effects matter. We now recap the intuition and discuss implications for competition policy.

In our model, a politician decides whether to approve a pro-competitive reform, which benefits consumers but reduces incumbent firms' profits. Firms can bribe the politician to block it. Before lobbying occurs, the two firms may merge, which increases incentives to lobby. Without the merger, blocking the reform is less profitable

due to price competition; with the merger, lobbying rises. The merger makes the reform less likely to pass but improves the voter's ability to screen politicians, as Bad politicians accepting bribes are more likely to be defeated. This trade-off complements conventional merger control, which considers only price effects.

The traditional consumer welfare standard is too weak whenever voters' screening benefits do not offset the loss in consumer surplus. Two cases arise. First, lobbying may occur regardless of the merger. Then, some corrupt politicians may prevent the reform, requiring stricter merger standards than the consumer welfare benchmark. Here, the standard fails not because of increased political power but because politicians may not always act in consumers' interest. A stricter antitrust standard is needed to compensate for the probability that Bad politicians do not approve market reforms.

The second and more interesting case arises when lobbying occurs only if the merger is approved. Now, the merger allows firms to influence Bad politicians to block the reform. It may even be optimal to reject mergers that would reduce prices, because rejecting the merger ensures the reform passes and expected prices are lower (Figure 4.1). A very strict standard is then optimal: high efficiency gains are required for merger approval, anticipating that the merger itself triggers lobbying and blocks pro-competitive reforms.

As the benefits from screening politicians increase, the welfare performance of the lobbying equilibrium improves. The merger-induced lobbying allows voters to defeat corrupt politicians with higher probability. When screening benefits are large, higher post-merger price increases may be tolerated to remove Bad politicians, making the consumer welfare standard potentially too strict.

When, in practice, can this be the case? One may want to approve anti-competitive mergers and bear the cost of bribery today in order to avoid re-electing Bad politicians tomorrow. Three conditions then need to be satisfied. First, the voters' time discount factor should not be too low. Second, there must be welfare gains from avoiding regulations detrimental to consumers/voters in *other* parts or sectors of the economy.

Suppose instead that the only regulation that corrupt politicians can implement to the detriment of consumers/voters is the one limiting the extent of competition within the narrowly defined market (i.e., the policy a). Then, it can never be optimal to suffer the policy for sure today in order to avoid it happening with some probability tomorrow.²⁹ Third, future regulation should not be too attractive for Bad politicians; otherwise, the Separating equilibrium disappears. Absent out-of-market externalities, the consumer welfare standard is always too lenient.

The consumer welfare standard is also too lenient when the incumbent is unpopular. In Footnote 11, we discuss the case where voters' prior beliefs favor the challenger over the incumbent. In this case, a Bad politician who behaves well and implements the reform can never be re-elected. Since pooling on the "good" action yields no electoral benefits, the Bad politician always accepts bribes. Whenever firms find it profitable to lobby, the equilibrium is thus always Separating, and the consumer welfare standard is always too lenient, essentially as in the middle case of Proposition 2.³⁰

Implications for Lobbying Regulation Our model shows that the stringency of the lobbying and merger regulations can be complements or substitutes from a social welfare perspective, depending on the benefits from screening. Suppose lobbying legislation is strict, so that the cost of lobbying is high enough that firms lobby if and only if the merger is approved. When screening benefits are low, merger policy and lobbying regulation are complements. The optimal merger policy should then be strict, as the lobbying equilibrium is inefficient for consumers/voters. The additional political power unlocked by the merger is used to influence the incumbent politician and block pro-competitive reform, and improved screening does not offset this effect. By contrast, when screening is efficient, lobbying raises welfare and merger policy can

²⁹We show this formally in Section C.2 in a two-period model.

³⁰When firms do not find it profitable to lobby and the voter's prior favors the challenger over the incumbent, there is a potential multiplicity of equilibria, as the politician would be indifferent between passing the pro-competitive reform or not. By the refinement introduced in Section 3.2, we select the voter's preferred equilibrium, whereby the Bad politician blocks the reform regardless. Therefore, even in this case, the consumer welfare standard is too lenient.

be weak, allowing mergers that induce voter learning, which has high social value. In this case, optimal merger policy and lobbying regulation are substitutes.

Implications for Competition Policy Should antitrust authorities consider non-market dimensions in their analysis? In principle, they have the statutory powers in merger investigations to request internal documents from the merging parties, which may include information on lobbying activities. If such evidence emerges, it could be used to subject the transaction to closer scrutiny.

However, we do not think this is feasible in many instances. [Tirole \[2022\]](#) warns about the risks of mission creep: expanding authorities' mandates to social and political goals can increase interference, weaken independence, reduce accountability, and dilute focus. Our model is silent on these organizational concerns, as we assume a benevolent authority without agency problems. While this is a natural avenue for future research, we can nonetheless draw implications for competition policy.

Real-world policy involves several dimensions of uncertainty not captured by our model, notably the burden and the standard of proof. The standard of proof is the degree to which a party must prove its case; the burden of proof is the obligation to meet that standard. Our results can be interpreted through this lens.

Currently, the burden of proof for efficiencies rests with firms, which is reasonable given their superior information. Our contribution instead speaks to the standard of proof. A strict optimal merger policy corresponds to a high standard for efficiency claims, with the strictest case being a structural presumption against mergers. Antitrust authorities can retain a consumer-centric mandate even in the presence of political effects. A stricter merger policy may be warranted because mergers can harm consumer welfare not only through market power but also through political channels. Our model thus provides conditions supporting stronger (rebuttable) structural presumptions in merger assessment, in line with the spirit of the 2023 revision of the US merger guidelines.³¹

³¹Source: [Department of Justice](#). See [Lancieri and Valletti \[2024\]](#).

5 Concluding Remarks

Zingales [2017] draws attention to the risk of a “Medici vicious circle,” in which economic and political power reinforce each other. Large firms can shape the rules of the markets in which they operate. A central institutional question is therefore how society should anticipate this process when designing enforcement regimes: how should policy account for the linkage between economic and political power?

In this paper, we study how antitrust enforcement should respond to the presence of political power, combining a simple model of mergers from industrial organization with a political agency model with firm lobbying. Our framework highlights a novel trade-off associated with mergers. By increasing firms’ political influence, mergers can harm consumers by weakening pro-competitive policies. At the same time, the increased political power of firms can improve voters’ ability to select un-corruptible politicians. We discuss how this trade-off interacts with standard competitive effects in merger assessment. Taken together, our results show that the consumer welfare standard is suboptimal once mergers’ political economy effects are taken into account.

Our findings rely on a highly stylized model, but they are robust to various extensions and robustness checks, as we show in the Appendix. Future work could extend our analysis to richer models of lobbying and voting. Exploring these directions will yield further insights into the interplay between market power and political power, and into the institutional design of competition policy.

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Appendix

A Proofs and Derivations of the Baseline Model

The solution of the model can be replicated by downloading the attached [Mathematica code](#).

A.1 Equilibrium Prices

Suppose the two firms have not merged ($m = 0$). Operating profits are as follows:

$$\hat{\pi}_i(p_i, p_j, p_3) = \frac{(p_i - c)(1 - \gamma - p_i(\gamma + 1) + (\gamma(p_j + p_3)))}{(1 - \gamma)(2\gamma + 1)}. \quad (\text{A.1})$$

The FOCs imply:

$$\begin{aligned} \frac{\partial \hat{\pi}_i(p_i, p_j, p_3)}{\partial p_i} &= \frac{1 + c - 2p_i + \gamma(c - 2p_i + p_j + p_3 - 1)}{1 - 2\gamma^2 + \gamma} = 0 \Rightarrow \\ p_i(p_j, p_3) &= \frac{\gamma(c + p_j + p_3 - 1) + c + 1}{2(\gamma + 1)}. \end{aligned} \quad (\text{A.2})$$

Intersecting firms' best response functions, we get:

$$p_i^{m=0}(p_3) = \frac{\gamma(c + p_3 - 1) + c + 1}{\gamma + 2}. \quad (\text{A.3})$$

It is easy to see that (A.3) is increasing in p_3 (a). Substituting (A.3) into demand functions and profits, we get:

$$\begin{aligned} d_i(p_i^{m=0}(p_3), p_j^{m=0}(p_3), p_3) &= \frac{(\gamma + 1)(1 - c - \gamma + \gamma p_3)}{(1 - \gamma)(\gamma + 2)(2\gamma + 1)} \\ \hat{\pi}_i(p_i^{m=0}(p_3), p_j^{m=0}(p_3), p_3) &= \frac{(\gamma + 1)(1 - c - \gamma + \gamma p_3)^2}{(1 - \gamma)(\gamma + 2)^2(2\gamma + 1)}, \end{aligned} \quad (\text{A.4})$$

with

$$\begin{aligned} \frac{\partial d_i \left(p_i^{m=0}(p_3), p_j^{m=0}(p_3), p_3 \right)}{\partial p_3} &> 0, \\ \frac{\partial \hat{\pi}_i \left(p_i^{m=0}(p_3), p_j^{m=0}(p_3), p_3 \right)}{\partial p_3} &> 0. \end{aligned} \quad (\text{A.5})$$

Note that demands are non-negative whenever $p_3 > \frac{c+\gamma-1}{\gamma}$. Therefore, in what follows, we assume $\gamma < 1 - c$.

Therefore, profits of firm $i \in \{1, 2\}$ increase in p_3 via two channels. First, an increase in the price of good 3 increases the demand of good i . Second, the higher the price of good 3, the milder the competition, and the higher the price that firm i can set.

If the merger is approved ($m = 1$), the merged firm solves:

$$\max_{p_1, p_2 \geq 0} \hat{\pi}_1(p_1, p_2, p_3) + \hat{\pi}_2(p_2, p_1, p_3). \quad (\text{A.6})$$

By FOCs, optimal prices are:

$$p_i^{m=1}(p_3) = \frac{1}{2}(c\mu + \gamma(p_3 - 1) + 1), \quad (\text{A.7})$$

with

$$\begin{aligned} d_i \left(p_i^{m=1}(p_3), p_j^{m=1}(p_3), p_3 \right) &= \frac{c\mu + \gamma - \gamma p_3 - 1}{4\gamma^2 - 2\gamma - 2}, \\ \hat{\pi}_1 \left(p_1^{m=1}(p_3), p_2^{m=1}(p_3), p_3 \right) + \hat{\pi}_2 \left(p_2^{m=1}(p_3), p_1^{m=1}(p_3), p_3 \right) &= \\ \frac{(c\mu + \gamma + \gamma(-p_3) - 1)^2}{2(1 - \gamma)(2\gamma + 1)}. \end{aligned} \quad (\text{A.8})$$

The comparative statics of (A.8) is analogous to (A.5).

We now compare prices (A.3), (A.7). The merger decreases prices if and only if

efficiencies are high enough:

$$p_i^{m=1}(p_3) < p_i^{m=0}(p_3) \Leftrightarrow \mu < \frac{2c(\gamma + 1) - \gamma(1 - \gamma + \gamma p_3)}{c(\gamma + 2)} := \bar{\mu}(p_3) . \quad (\text{A.9})$$

A.2 Returns from Protection

If $m = 0$, firms' return from protection is:

$$\begin{aligned} \Delta\pi_i(0) = & \\ \hat{\pi}_i(p_i^{m=0}(\eta), p_j^{m=0}(\eta), \eta) - \hat{\pi}_i(p_i^{m=0}(0), p_j^{m=0}(0), 0) = & \quad (\text{A.10}) \\ \frac{\gamma(\gamma + 1)\eta(2(1 - c - \gamma) + \gamma\eta)}{(1 - \gamma)(\gamma + 2)^2(2\gamma + 1)} . \end{aligned}$$

If $m = 1$, firms' return from protection is:

$$\begin{aligned} \Delta\pi_i(1) = & \\ \hat{\pi}_i(p_i^{m=1}(\eta), p_j^{m=1}(\eta), \eta) - \hat{\pi}_i(p_i^{m=1}(0), p_j^{m=1}(0), 0) = & \quad (\text{A.11}) \\ \frac{\gamma\eta(\gamma(\eta - 2) - 2c\mu + 2)}{4(1 - 2\gamma^2 + \gamma)} . \end{aligned}$$

It is easy to see that $\Delta\pi(1) > \Delta\pi(0) \geq 0$ always.

A.3 Merger Policy

We start by writing down the authority's payoff across the different equilibria. Let $m = 0$. In a Pooling equilibrium :

$$\begin{aligned} W_{Pool}(0) = & \\ \frac{1}{4} \left(c^2 - 2c - 4(1 - q)\phi + 3 \right) + & \quad (\text{A.12}) \\ + \frac{\gamma(6\gamma^3 + c^2(2\gamma^2 + 7\gamma + 7)\gamma + 7\gamma^2 + c(-4\gamma^3 - 6\gamma^2 + 2\gamma + 8) - 5\gamma - 8)}{4(1 - \gamma)(\gamma + 2)^2(2\gamma + 1)} . \end{aligned}$$

Let $m = 1$. In a Pooling equilibrium :

$$W_{Pool}(1) = \frac{1}{4} \left(c^2 \mu^2 - 2c\mu + 4(q-1)\phi + 3 \right) - \frac{\gamma (\gamma (-2c^2 \mu^2 + 4c\mu - 3) + c^2 \mu^2 - 4c\mu + 3)}{4(1-\gamma)(2\gamma+1)} . \quad (A.13)$$

Let $m = 0$. In a Separating equilibrium :

$$W_{Sep}(0) = A\eta^2 + B\eta + C , \quad (A.14)$$

where

$$\begin{aligned} C &= \frac{2c^2(\gamma+1)^2 + 4c(\gamma-1)(\gamma+1)^2 + (\gamma-1)(-7\gamma^2 - 14\gamma + (\gamma+2)^2(2\gamma+1)q - 6)}{2(1-\gamma)(\gamma+2)^2(2\gamma+1)} + \\ &+ \frac{q}{2} - (q-1)\phi(qx-1) ; \\ A &= \frac{(4-\gamma(3\gamma^2+\gamma-8))(1-q)}{2(1-\gamma)(\gamma+2)^2(2\gamma+1)} ; \\ B &= \frac{(4c\gamma(\gamma+1)^2(q-1) - 2(\gamma-1)(5\gamma(\gamma+2)+4)(q-1))}{2(1-\gamma)(\gamma+2)^2(2\gamma+1)} . \end{aligned} \quad (A.15)$$

Let $m = 1$. In a Separating equilibrium :

$$W_{Sep}(1) = D\mu^2 + E\mu + F , \quad (A.16)$$

where

$$F =$$

$$\frac{1}{4(\gamma - 1)(2\gamma + 1)} \left[4\gamma\phi + 2\gamma\eta^2(q - 1) - 2\gamma\eta(q - 1) + 2(\eta - 2)\eta(q - 1) + \gamma^2(-3(\eta - 2)\eta(q - 1) - 8(q - 1)\phi(qx - 1) + 3) + 4\gamma q\phi((q - 1)x - 1) + 4q\phi((q - 1)x - 1) + 4\phi - 3 \right]$$

$$D =$$

$$\frac{c^2}{4(1 - \gamma)(2\gamma + 1)};$$

$$E =$$

$$\frac{(-2\gamma c - 2\gamma c\eta(q - 1) + 2c)}{4(\gamma - 1)(2\gamma + 1)}.$$

(A.17)

We now obtain the critical thresholds $\tilde{\mu}, \bar{\mu}(0), \hat{\mu}, \bar{\mu}(\eta)$. These thresholds are obtained from finding the admissible roots of a quadratic equation in μ . Specifically, for all $h, l \in \{Sep, Pool\}$, we solve

$$W_k(1) - W_l(0) = W\mu^2 + S\mu + T, \quad (A.18)$$

where W, S, T are constant in μ and $W > 0$ always. Findings are reported next.

$$\mu \leq \hat{\mu} \Rightarrow W_{Sep}(1) \geq W_{Sep}(0), \text{ where}$$

$$\hat{\mu} =$$

$$\frac{(\gamma + 2)(1 - \gamma + \gamma\eta(1 - q)) - \sqrt{G\eta^2 + H\eta + I}}{c(\gamma + 2)} \text{ and}$$

(A.19)

$$G = \gamma^2(1 - q) \left(4(\gamma + 1)^2 - (\gamma + 2)^2 q \right)$$

$$H = 8\gamma(\gamma + 1)^2(1 - q)(1 - c - \gamma)$$

$$I = 4(\gamma + 1)^2(c + \gamma - 1)^2;$$

$\mu \leq \tilde{\mu} \Rightarrow W_{Sep}(1) \geq W_{Pool}(0)$, where

$\tilde{\mu} =$

$$\frac{(\gamma + 2)(1 - \gamma + \gamma\eta(1 - q)) - \sqrt{L\gamma^4 + M\gamma^3 + N\gamma^2 + Q\gamma + R}}{c(\gamma + 2)} \text{ and}$$

$$L = (1 - q)(8qx\phi - \eta(8 - \eta(q - 4))) + 4 \quad (A.20)$$

$$M = 8c - 2(1 - q) (\eta(\eta(2q - 7) + 14) - 14qx\phi)$$

$$N = 4c(c + 2) - 2(1 - q) (\eta(\eta(2q - 3) + 6) - 6qx\phi) - 8$$

$$Q = 8((c - 1)c - 2(1 - q)((\eta - 2)\eta + 2qx\phi))$$

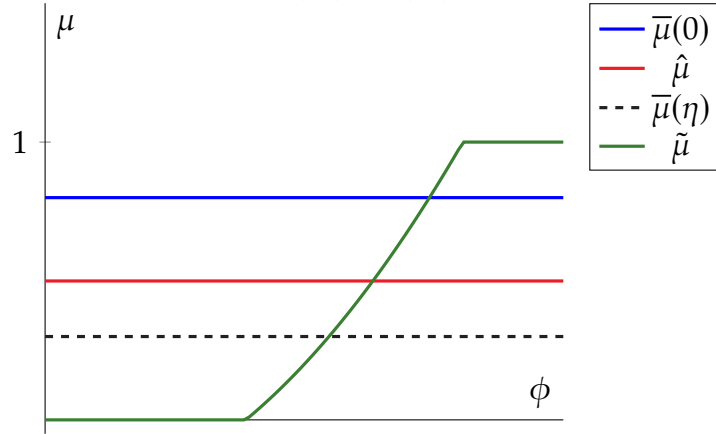
$$R = 4(c - 2)c - 8(1 - q)((\eta - 2)\eta + 2qx\phi) + 4 ;$$

$\mu \leq \bar{\mu}(0) \Rightarrow W_{Pool}(1) \geq W_{Pool}(0)$, where

$$\bar{\mu}(0) = \frac{2c(\gamma + 1) + (\gamma - 1)\gamma}{c(\gamma + 2)} . \quad (A.21)$$

Additional details are available in the attached [Mathematica code](#). In Figure A.1, we plot the different thresholds as a function of ϕ .

Figure A.1: Thresholds $\tilde{\mu}, \bar{\mu}(0), \hat{\mu}, \bar{\mu}(\eta)$ as a function of ϕ



A.4 Proof that $\tilde{\mu} \geq \bar{\mu}(0)$ (when ϕ is high) and $\tilde{\mu} < \bar{\mu}(\eta)$ (when ϕ is low)

The authority approves the merger if and only if:

$$\begin{aligned}
 W_{Sep}(1) - W_{Pool}(0) &\geq 0 = \\
 q \left[u \left(d^* \left(p^{m=1}(0) \right) \right) \right] &+ \\
 (1-q) \left[u \left(d^* \left(p^{m=1}(\eta) \right) \right) + (1-x)(-\phi) + x(1-q)(-\phi) \right] &+ \\
 - u \left(d^* \left(p^{m=0}(0) \right) \right) - (1-q)(-\phi) &\geq 0.
 \end{aligned} \tag{A.22}$$

Let us define $s(\phi)$ as the expected benefit for the voter in the Separating equilibrium :

$$s(\phi) = -\phi \Delta P(\theta^* = \text{Bad}), \tag{A.23}$$

where

$$\begin{aligned}
 \Delta P(\theta^* = \text{Bad}) &= \\
 [(1-q)(x(1-q) + (1-x))] - (1-q) &= \\
 - (1-q)qx &\leq 0
 \end{aligned} \tag{A.24}$$

is the difference in the probability of election of a Bad P between the two equilibria. Since $s(\phi) \geq 0$ always, the voter prefers the Separating equilibrium. Moreover, $s' > 0$ and $s(0) = 0$.

Now, we rewrite (A.22) as follows:

$$\begin{aligned}
 W_{Sep}(1) - W_{Pool}(0) &\geq 0 = \\
 \left[qu \left(d^* \left(p^{m=1}(0) \right) \right) + (1-q) u \left(d^* \left(p^{m=1}(\eta) \right) \right) - u \left(d^* \left(p^{m=0}(0) \right) \right) \right] &+ s(\phi) \geq 0
 \end{aligned} \tag{A.25}$$

(A.25) allows us to isolate the effects of the merger on the consumer's and the voter's payoffs.

Let us assume $\phi = 0$ and consider the consumer's perspective. If the merger is approved, the reform is implemented with probability q , while if the merger is not approved, the reform is implemented with probability 1. Therefore, a very high level of efficiencies (lower μ) is needed for the merger to be pro-competitive. In particular, if $\phi = 0$, $\tilde{\mu} < \bar{\mu}(0)$ necessarily.

We can see this by contradiction. Assume $\phi = 0$ and $\tilde{\mu} \geq \bar{\mu}(0)$. For the merger to be pro-competitive, it must decrease prices. Then,

$$\begin{aligned} \mu \in [\bar{\mu}(0), \tilde{\mu}) &\Rightarrow \\ qp_i^{m=1}(0) + (1-q)p_i^{m=1}(\eta) &\leq p_i^{m=0}(0) \Rightarrow \\ p_i^{m=1}(0) &< p_i^{m=0}(0). \end{aligned} \tag{A.26}$$

which contradicts (4.4). In words, if $\mu \in (\bar{\mu}(0), \tilde{\mu})$ and the merger is pro-competitive when the reform is implemented with probability < 1 , then it must also be pro-competitive when the reform is implemented with probability 1, which we know it is not the case. Finally, since (A.25) is increasing in ϕ , then also $\tilde{\mu}$ increases in ϕ .³²

By the same logic, one can see that $\tilde{\mu}$ can be lower than $\bar{\mu}(\eta)$. To see this, consider the extreme case when $q \rightarrow 0$ (almost all politicians are Bad) and $\phi = 0$. The authority approves the merger if and only if

$$\begin{aligned} u\left(d^*\left(p^{m=1}(\eta)\right)\right) &> u\left(d^*\left(p^{m=0}(0)\right)\right) \Rightarrow \\ p^{m=1}(\eta) &< p^{m=0}(0). \end{aligned} \tag{A.27}$$

Recall that

$$\mu \leq \bar{\mu}(\eta) \Rightarrow p^{m=1}(\eta) < p^{m=0}(\eta). \tag{A.28}$$

Since

$$p^{m=0}(0) < p^{m=0}(\eta), \tag{A.29}$$

³²Suppose $\phi \rightarrow \infty$, then even if the merger does not bring efficiencies ($\mu = 1$), it is implemented. By monotonicity, it exists some $\tilde{\phi} < \infty$ such that $\tilde{\mu} = \bar{\mu}(0)$.

a higher level of efficiencies is required for (A.27) to hold:

$$\tilde{\mu} < \bar{\mu}(\eta) . \quad (\text{A.30})$$

We show next some numerical examples of the relationship between $\tilde{\mu}$, $\bar{\mu}(\eta)$, $\bar{\mu}(0)$, $\hat{\mu}$, and we comment on some limiting cases.

Table A.1 shows that as we increase ϕ , $\tilde{\mu}$ becomes larger than $\bar{\mu}(0)$.

Table A.2 shows that when $q = 1$ (all politicians are Good), the optimal merger policy always coincides with the consumer welfare standard $\bar{\mu}(0)$.

Table A.3 shows that when $q \rightarrow 0$ (almost all politicians are Bad), $\hat{\mu} = \bar{\mu}(\eta)$. In this case, $\tilde{\mu} = 0$. No merger is approved when allowing the merger creates a Separating equilibrium, as this Separating equilibrium is very inefficient from the consumer's perspective.³³

ϕ	$\tilde{\mu}$	$\bar{\mu}(0)$	$\hat{\mu}$	$\bar{\mu}(\eta)$
0.3	0.000	0.367	0.223	0.117
0.5	0.000	0.367	0.223	0.117
0.8	0.135	0.367	0.223	0.117
1	0.952	0.367	0.223	0.117

Table A.1: Optimal merger policy when $\eta = 0.3; \gamma = 0.5; x = 0.5; q = 0.5; c = 0.12$

ϕ	$\tilde{\mu}$	$\bar{\mu}(0)$	$\hat{\mu}$	$\bar{\mu}(\eta)$
0.3	0.367	0.367	0.367	0.117
0.5	0.367	0.367	0.367	0.117
0.8	0.367	0.367	0.367	0.117
1	0.367	0.367	0.367	0.117

Table A.2: Optimal merger policy when $\eta = 0.3; \gamma = 0.5; x = 0.5; q = 1; c = 0.12$

³³In this case, since all politicians are Bad, the voter pays the penalty $-\phi$ no matter if the equilibrium is Pooling or Separating. When there are no Good politicians, screening has no value. Note that we assume that there is always a strictly positive probability that politicians are Good ($q > 0$). If this were not the case ($q = 0$), Pooling on the "good" action (rejecting bribes and approving the reform) would never be optimal for a Bad politician because they would always be re-elected even after accepting bribes.

ϕ	$\tilde{\mu}$	$\bar{\mu}(0)$	$\hat{\mu}$	$\bar{\mu}(\eta)$
0.3	0.000	0.367	0.117	0.117
0.5	0.000	0.367	0.117	0.117
0.8	0.000	0.367	0.117	0.117
1	0.000	0.367	0.117	0.117

Table A.3: Optimal merger policy when $\eta = 0.3; \gamma = 0.5; x = 0.5; q = 0; c = 0.12$

B Generalization of Proposition 1

In this Section, we show that Proposition 1 follows from the supermodularity of the Bertrand game, and so it easily extends to any supermodular game. The concepts used in this Section are standard and come from Milgrom and Shannon [1994], Topkis [1998], and Levin [2003].

Let us define the following game. Let $i \in \{1, 2\}$ index the two firms. Each firm's action is $a_i \in A_i$, where A_i is the action space, which we assume to be a compact set. Let $\pi_i(a_i, a_j, a)$ be the payoff of firm i , where $a \in \{0, 1\}$ is the politician's action, which we interpret as in the baseline model, so that $\pi_i(a_i, a_j, a)$ is decreasing in a . Assume P chooses a before firms choose their actions. Then, we can interpret a as a parameter. We assume that $\pi_i(a_i, a_j, a)$ is continuous.

If $m = 1$, firms' choose a_i, a_j cooperatively. If $m = 0$, they choose a_i, a_j simultaneously and independently. Our aim is to show that each firm's marginal benefit from reducing a increases after a merger ($m = 1$).

Before introducing the notion of supermodularity, we define two additional elements. First, for expositional convenience, let us introduce the following notation: $\bar{a} = -a$, so that $\pi_i(a_i, a_j, \bar{a})$ increases in $\bar{a}$. Second, we introduce the property of Increasing Differences (ID). A function $f(x, t)$ satisfies ID in (x, t) if and only if, for all $x \in X, t \in T$, where X, T are two (partially) ordered sets, such that $x' \geq x, t' \geq t$, the following property holds:

$$f(x', t') - f(x, t') \geq f(x', t) - f(x, t) . \quad (\text{B.1})$$

Condition (B.1) implies that the marginal value from increasing x is higher when t is higher.

We assume that the game described above is a supermodular game with positive spillovers indexed in $\bar{a}$ (Levin [2003]), that is,

- (a) For all i, j , the function $\pi_i(a_i, a_j, \bar{a})$ satisfies ID in (a_i, a_j) ;
- (b) For all i , the function $\pi_i(a_i, a_j, \bar{a})$ satisfies ID in $(a_i, \bar{a})$;
- (c) For all i, j , the function $\pi_i(a_i, a_j, \bar{a})$ increases in a_j .

Let us define $(a_i^{m=1}, a_j^{m=1})$ as the optimal actions for the two firms if $m = 1$.³⁴ If $m = 0$, by the supermodularity of the game, there is at least one Nash Equilibrium in pure strategies. Let $(a_i^{m=0}, a_j^{m=0})$ be the Nash Equilibrium actions of the game if $m = 0$.³⁵

By (c):

$$(a_i^{m=1}, a_j^{m=1}) \geq (a_i^{m=0}, a_j^{m=0}), \quad (\text{B.2})$$

which implies:

$$\forall \bar{a}, \pi_i(a_i^{m=1}, a_j^{m=1}, \bar{a} = 0) - \pi_i(a_i^{m=0}, a_j^{m=0}, \bar{a} = 0) \geq 0. \quad (\text{B.3})$$

Because of pair-wise ID between any pair of its arguments, the function π_i is supermodular.³⁶ Therefore, increasing both actions a_i, a_j has an higher marginal return when $\bar{a}$ is higher:

$$\begin{aligned} & \pi_i(a_i^{m=1}, a_j^{m=1}, \bar{a} = 0) - \pi_i(a_i^{m=0}, a_j^{m=0}, \bar{a} = 0) \geq \\ & \pi_i(a_i^{m=1}, a_j^{m=1}, \bar{a} = -1) - \pi_i(a_i^{m=0}, a_j^{m=0}, \bar{a} = -1) \end{aligned} \quad (\text{B.4})$$

³⁴These maximizers exist by the continuity of payoff functions and the compactness of action spaces.

³⁵In case of a multiplicity of equilibria, we select, without loss of generality, the Pareto un-dominated equilibrium, which exists by (c) (Levin [2003]).

³⁶To apply the properties of supermodular functions, we need to impose the assumption that the set $A_i \times A_j \times \{0, -1\}$ is a lattice.

and by the symmetry of ID:

$$\begin{aligned} \pi_i \left(a_i^{m=1}, a_j^{m=1}, \bar{a} = 0 \right) - \pi_i \left(a_i^{m=1}, a_j^{m=1}, \bar{a} = -1 \right) &\geq \\ \pi_i \left(a_i^{m=0}, a_j^{m=0}, \bar{a} = 0 \right) - \pi_i \left(a_i^{m=0}, a_j^{m=0}, \bar{a} = -1 \right) &\geq 0 \end{aligned} \quad (\text{B.5})$$

Intuitively, because of a merger, firms set actions cooperatively and take positive spillovers into account. Then, they choose higher actions. If the marginal value of increasing actions a_i, a_j increases as $\bar{a}$ increases (a decreases), then, by symmetry, the marginal value of increasing $\bar{a}$ (decreasing a) increases when actions are higher.

Note that the Bertrand game is a supermodular game. The Cournot duopoly is a supermodular game if one player's strategy set is given the reverse of its usual order (Levin [2003]).³⁷ Note also that this proof does not require any assumption on the degree, concavity, or differentiability of demand and profit functions. If a_i, a_j are interpreted as prices, then (B.5) implies³⁸

$$\forall i, a_i^{m=1}(\bar{a} = 0) - a_i^{m=0}(\bar{a} = 0) > a_i^{m=1}(\bar{a} = -1) - a_i^{m=0}(\bar{a} = -1), \quad (\text{B.6})$$

which indicates that the welfare analysis performed in Section 4 would easily extend to this more general model.³⁹

If the sign of (B.5) is inverted, then a merger decreases firms' incentives to lobby. We have shown that this can never be the case if the market interaction is a game with spillovers and supermodular payoffs.

³⁷For more examples, see Levin [2003].

³⁸ Under this interpretation, (B.6) implies that, after a merger, each firm i increases prices more when the politician blocks the pro-competitive reform (i.e., when the price of good 3 is higher). Recall that, in this section, the two firms' profit function is increasing in $\bar{a}$, and that in the baseline model the firms' profit function is increasing in the price of good 3.

³⁹We can see this by looking at Appendix A.4, where we derive the properties of the optimal merger policy relying only on the impact of a and m on prices and on ϕ .

C Supplementary Material: Extensions

In the main text, we presented a simple model highlighting the political effects of mergers and how these effects interplay with market power considerations in a merger assessment. We now extend the model to discuss the robustness of our findings. In particular, we first consider an alternative bargaining structure where firms offer a mechanism to P. Then, we consider an alternative model setup with two periods and endogenous t , ϕ , and V .

C.1 Firms Offer the Mechanism

Let us consider a modified version of the timing introduced in Section 2. In [Stage 2](#), both firms submit a TIOLI offer to P: $l_i(a) \in \{0, 1\}$. If $m = 1$, the two (merged) firms coordinate their offers. If $m = 0$, the two firms submit their offers simultaneously and independently. In [Stage 3](#), P observes both offers and decides on the implementation of the reform.

How does this alternative bargaining structure affect the nature of the lobbying equilibrium? Suppose $m = 1$. Suppose further that

$$(1 - x)V + t \geq V \Rightarrow t \geq Vx . \quad (\text{C.1})$$

Condition [\(C.1\)](#) implies that a single bribe of value t is sufficient to make $a = 0$ IC for P. In the baseline model, even if [\(C.1\)](#) holds, P requires the payment of two bribes for a total amount of $2t$ to set $a = 0$ because they have the power to commit to that, and of course, $2t > t$. In this model, if [\(C.1\)](#) holds, the optimal mechanism for the merged firms is:

$$\begin{cases} l_i^*(0) = 1 \\ l_j^*(0) = 0 . \end{cases} \quad (\text{C.2})$$

Since the merged firms know that P is willing to set $a = 0$ in exchange for t , there is no incentive to offer more and pay $2t$. In this case, only one firm lobbies, and P does

not implement the policy. This equilibrium exists if and only if $t \in [Vx, \Delta\pi(1)]$. If $t \in [\frac{Vx}{2}, Vx)$, the merged firms know that they need to pay a total of $2t$ in order to make $a = 0$ IC for P, as in the baseline model.

Now, suppose that $m = 0$. As in the previous case, if $t \in [\frac{Vx}{2}, Vx)$, both firms lobby in equilibrium. When $t \in [Vx, \Delta\pi(0)]$, there exist two equilibria in pure strategies. In the first equilibrium, firm i commits to the offer $l_i^*(0) = 1$ and firm j commits to $l_j^*(0) = 0$, and P chooses $a^* = 0$. In the second equilibrium, firm j lobbies, and firm i does not.⁴⁰

The multiplicity of equilibria generates a coordination problem between the two firms. To illustrate this, let us consider the following profile of mixed strategies. Suppose that firm 1 makes the offer $l_1(0) = 1$ with probability $k \in (0, 1)$ and commits not to lobby with probability $1 - k$. Let $\tilde{l}_1$ denote this mixed strategy. If firm 2 makes the offer $l_2(0) = 1$, its expected payoff is:

$$\mathbb{E} [l_2(0) = 1, \tilde{l}_1] = \hat{\pi}_2(p^{m=0}(\eta)) - t. \quad (\text{C.3})$$

If firm 2 commits to $l_2 = 0$, its expected payoff is:

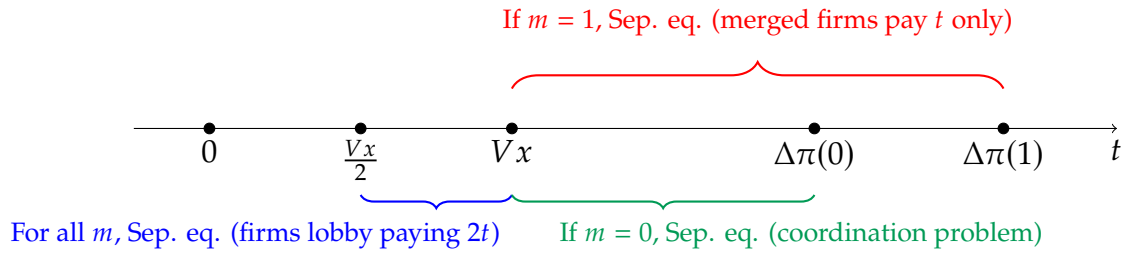
$$\mathbb{E} [l_2 = 0, \tilde{l}_1] = k \left(\hat{\pi}_2(p^{m=0}(\eta)) \right) + (1 - k) \hat{\pi}_2(p^{m=0}(0)). \quad (\text{C.4})$$

It follows that if $k = \frac{\Delta\pi(0)-t}{\Delta\pi(0)} := k^*$, there exists an equilibrium in mixed strategies where both firms lobby with probability k^* . In this equilibrium, with probability $(1 - k^*)^2$, neither firm lobbies and P does not implement the policy. This coordination failure arises from the incentive to free-ride on the competitor's lobbying effort.

Figure C.1 summarizes these results. By comparing Figures C.1 and 3.1, we can see that this alternative bargaining structure does not affect the likelihood that lobbying emerges in equilibrium, but only the extent of lobbying.

⁴⁰There is no equilibrium where both firms lobby because, given that firm i is lobbying, firm j does not want to lobby.

Figure C.1: Existence of Separating Equilibrium when firms have commitment power



Merger Assessment We now briefly discuss the welfare analysis of the merger. In particular, we aim to show that the welfare analysis of the merger is analogous to the baseline model.

In this section, firms commit to a TIOLI offer to the politician rather than vice-versa. We have shown that this alternative assumption affects lobbying at the "intensive margin" but not at the "extensive" one: it affects *how much* firms lobby, but not *when* they do it. Formally, there exists a parameter region, identified by $t \in [Vx, \Delta\pi(1)]$, where the two firms can induce the politician to block the pro-competitive reform while paying, in aggregate terms, only one bribe of value t . Notice that this is the only region where the equilibrium of this section differs from the equilibrium of the baseline model (see Figures C.1 and 3.1). Outside this region, the two models have the same solution, and thus the optimal merger policy in the two models is identical.⁴¹ In what follows, we then restrict our attention to the case of $t \in [Vx, \Delta\pi(1)]$.

Within this region, let us study separately the case of $m = 1$ (the merger is approved) and $m = 0$ (the merger is not approved). If there is a merger ($m = 1$), the two firms can always coordinate, and pay only one bribe of value t . This is enough to induce the Bad P to block the pro-competitive reform. The equilibrium is Separating,

⁴¹For instance, when $t > \Delta\pi(1)$, there is never lobbying in equilibrium, and the political equilibrium is always Pooling. When $t \in [\frac{Vx}{2}, Vx)$, there is a Separating equilibrium both if $m = 0$ and if $m = 1$. In both cases, each firm pays a bribe of value t to induce the Bad politician to block the pro-competitive reform. A total amount of $2t$ in lobbying money is paid by the industry, no matter if the merger is approved or not. If $t < \frac{Vx}{2}$, there is never lobbying in equilibrium because the Bad politician does not value bribes enough to give up re-election (with some probability). Also in this case, then, the equilibrium is always Pooling. In all these cases, this alternative model's solution is analogous to the solution of the baseline model.

and a total amount of t in lobbying money is paid by the two firms.

If there is no merger, two different cases arises. First, suppose that $t \in (\Delta\pi(0), \Delta\pi(1)]$. In this case, when $m = 0$, there is no lobbying. Therefore, in this region, the merger changes the nature of the political equilibrium, as it creates a Separating equilibrium. The optimal merger policy, in this case, coincides with the one derived for the baseline model (see Section 4). That is, $\mu^* = \tilde{\mu}$. The merger induces the two politicians' types to separate, and the antitrust authority balances the voter's gain from improved screening with the consumer's loss from the block of the pro-competitive reform.⁴² The only change with respect to the baseline model is that the merged entity needs to spend less in lobbying in order to obtain the same favor from the Bad P (the block of the pro-competitive reform). However, neither the voter nor the consumer are directly affected by the amount of lobbying. This only affects the distribution of surplus between P and the two firms. Since the antitrust authority cares only about the voter and the consumer (whose payoffs only depend on prices and on the probability that the pro-competitive reform is blocked), its problem is analogous to the one we solve in the baseline model.

Second, suppose that $t \in [Vx, \Delta\pi(0)]$. In this case, there is lobbying in equilibrium even if the merger does not go ahead. The equilibrium is Separating for all m . There is, however, a multiplicity of pure strategy equilibria, where one firm pays the bribe t , and the other does not; a situation we describe as a coordination problem. Therefore, we face the issue of equilibrium selection.

Let us start by focusing on pure strategies. Then, we select the equilibrium where firm 1 lobbies, and firm 2 does not lobby (the symmetric case is analogous). Suppose the antitrust authority anticipates that firms will play according to this equilibrium. In this case, the equilibrium is always Separating, no matter if the merger is approved or not. Unlike the baseline model, this equilibrium features lower bribes. But, as discussed above, this does not affect the antitrust authority's decision because it does not change neither prices nor the probability that the pro-competitive reform is

⁴²As we show in the main text, the solution of this trade-off depends on ϕ .

blocked. Therefore, $\mu^* = \hat{\mu}$. We can then state the following result.

Proposition C.1. *Suppose firms coordinate on a pure strategy lobbying equilibrium. The authority allows the merger ($m^* = 1$) if and only if $\mu \leq \mu^*$, where μ^* is characterized as in Proposition 2.*

Proof. The antitrust authority's payoff is the sum of the voter's and the consumer's. These depend only on prices and on the probability that the pro-competitive reform is blocked, and not on the amount of lobbying expenditures borne by the two firms. Since the nature of the equilibrium is identical to the baseline model, the welfare comparison faced by the authority is unchanged, and the thresholds in Proposition 2 apply directly. $\square$

The cases discussed are summarized in Figure C.2.

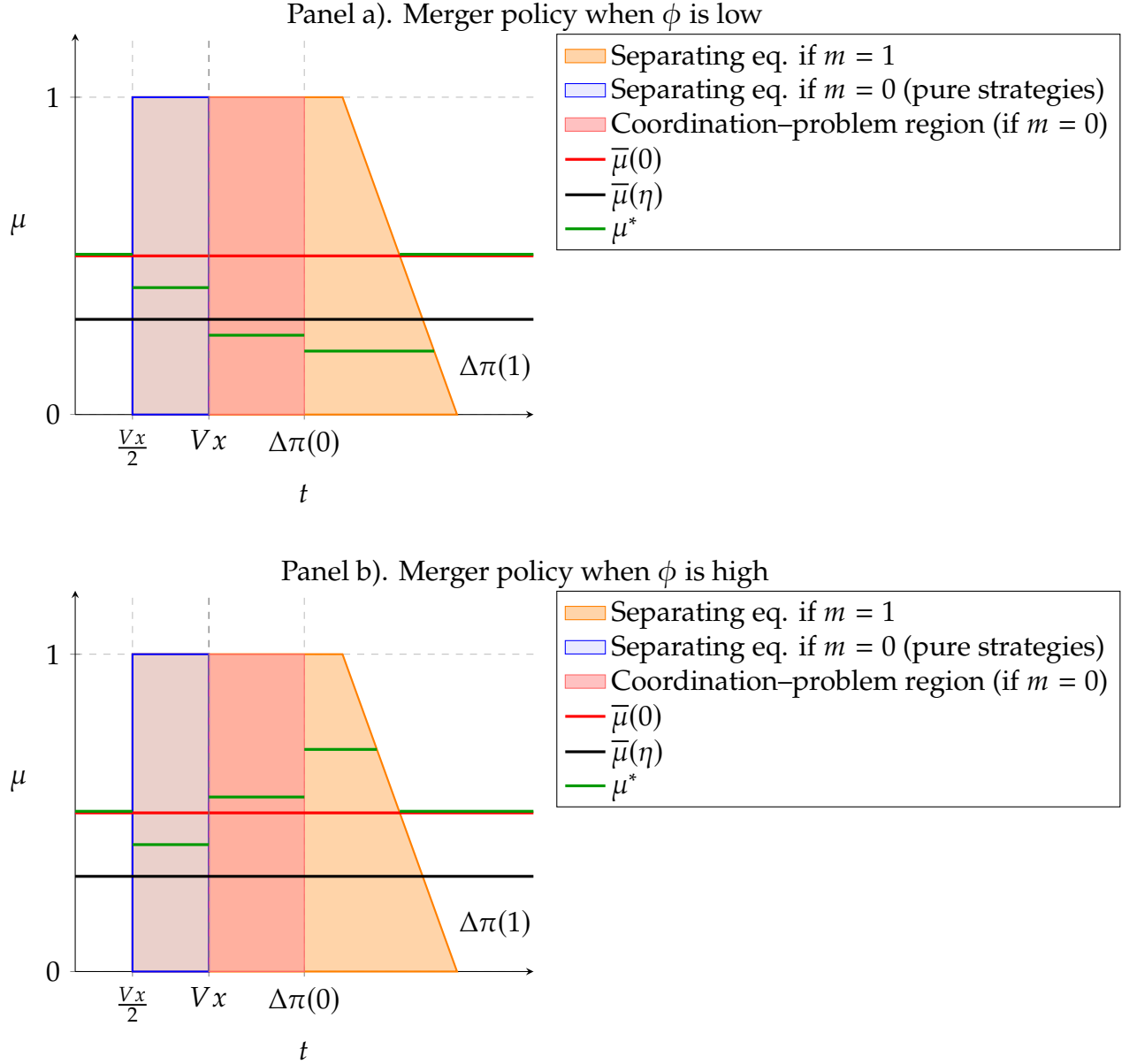
So far, we have proved that, restricting our attention to pure strategy equilibria, the main results from the baseline model extend to this alternative timing structure.

One may wonder, however, what happens if we select the mixed strategy equilibria we characterized above. How does the antitrust authority behave when it anticipates that each firm will lobby with probability $k^* < 1$? We now briefly discuss this case. The key observation is that the merger strictly increases the probability that a Separating equilibrium appears. Indeed, when the merger is not approved, there is a strictly positive probability (we denote it by: $\kappa := (1 - k^*)^2 > 0$) that neither of the two firms lobby. In this case, the Bad P does not receive any bribe, and it approves the pro-competitive reform. Then, if $m = 0$, the equilibrium is Pooling with probability κ , and Separating with probability $1 - \kappa$. If $m = 1$, the equilibrium is Separating with probability one. In this very specific case, the merger increases lobbying not because of the mechanism described by Proposition 1, but rather because it solves a coordination problem. However, from the antitrust authority's perspective, these two cases are analogous. Then, the optimal merger policy is qualitatively similar to the one described in Paragraph (c) of Section 4. That is, the optimal merger policy

approving the merger increases the probability that the two politician types separate. However, without the merger, separation already occurs with probability $1 - \kappa$, rather than with zero probability (as in the orange region, when $m = 0$). As a result, both the screening benefit and the lobbying costs generated by the merger are smaller in the red region than in the orange one. This is why, in Figure C.3, the optimal merger policy in the red region is “less extreme” than in the orange region. However, by comparing Figures C.2 and C.3, we can see that, no matter what equilibrium is selected, the main qualitative properties of Figure 4.1 are not affected by this alternative bargaining structure.

Remarks There are two main takeaways from this robustness check. First, for all m , when firms have the power of commitment on the offer to P, there exist equilibria where only one bribe worth t is paid. Firms can extract more surplus from P, as they can get protection ($a = 0$) at a lower price (t rather than $2t$). However, if the two firms do not merge ($m = 0$), the potential multiplicity of equilibria generates a coordination problem. The merger solves the coordination problem. Second, we have shown that our main welfare results are robust to this alternative bargaining structure.

Figure C.3: Optimal merger policy when firms commit to a TIOLI offer to P (selecting a mixed strategy equilibrium)



C.2 Endogenous t, ϕ, V in a Two-Period Model

In this Section, we introduce two key novelties. First, the political game is repeated twice, as in most political agency models. This allows us to endogenize the cost for consumers to re-elect a Bad P and the value of the office for a Bad P. Second, we allow P to specify t as part of the contract. We show that the main results of the baseline model extend to this more general setup. For the sake of simplicity, we assume perfect

monitoring ($x = 1$). Then, we can think of the consumer and the voter as the same player.

The timing is as follows. With a slight abuse of notation, we refer to the time between [Stage 1'](#) and [Stage 5'](#) as the First Period and to the time between [Stage 6'](#) and [Stage 8'](#) as the Second Period. As in the baseline model, firms 1, 2 merge if $m = 1$, and do not merge if $m = 0$.

Stage 1' Nature draws θ_P, θ_C .

Stage 2' P commits to a mechanism $a(l_1, l_2, t)$.

Stage 3' Firms observe the mechanism and choose whether to lobby ($l_i = 1$) or not ($l_i = 0$).

The reform a is implemented.

Stage 4' Firms observe $p_3(a)$ and simultaneously set prices p_1, p_2 . The consumer observes prices and chooses a consumption plan d_1, d_2, d_3 . First-period payoffs are realized.⁴⁴

Stage 5' The consumer chooses whether to re-elect P ($r = 1$) or not ($r = 0$).

Stage 6' The elected politician (with type θ^*) commits to a mechanism $a'(l'_1, l'_2, t')$.

Stage 7' Firms observe the mechanism and choose whether to lobby ($l'_i = 1$) or not ($l'_i = 0$).

The second-period reform a' is implemented.

Stage 8' Firms observe $p_3(a')$ and simultaneously set prices p'_1, p'_2 . The consumer observes prices and chooses a consumption plan d'_1, d'_2, d'_3 . Second-period payoffs are realized.

During the first period, players discount the future at a common discount factor $\delta \in (0, 1)$. The parameter δ can be interpreted as the exogenous probability that the game ends before the second period.

⁴⁴This assumption is not consequential. We would obtain analogous results by assuming that payoffs are realized at the end of the game.

The consumer's lifetime utility is then:

$$u(d_1, d_2, d_3) + \delta [u(d'_1, d'_2, d'_3) - s] , \quad (\text{C.5})$$

where $u(d_1, d_2, d_3)$ is defined as in (2.1) and

$$s = \begin{cases} \phi' & \text{if } \theta^* = \text{Bad} \\ 0 & \text{otherwise} , \end{cases} \quad (\text{C.6})$$

for some $\phi' \geq 0$. The parameter ϕ' captures the consumer's anticipated cost from appointing a Bad P, in addition to the market's partial equilibrium effects. If $\phi' = 0$, the consumer only cares about their utility within the market. If $\phi' > 0$, re-electing a Bad PM has a negative effect beyond the market. The lifetime payoff for a Bad P is:

$$V' + t(l_1 + l_2) + r\delta(V' + t'(l'_1 + l'_2)) . \quad (\text{C.7})$$

In this case, $V' \geq 0$ may be seen as the value of re-election, on top of bribes. For instance, V' may be the politician's salary.⁴⁵ As in the baseline model, the Good P has a behavioral type and always implements the policy. Firms' lifetime profits are:

$$\pi_i(p_i, d_i, l_i) + \delta(\pi_i(p'_i, d'_i, l'_i)) , \quad (\text{C.8})$$

where $\pi_i(p_i, d_i, l_i)$ is defined as in (2.4). To characterize the equilibrium, we proceed by Backward Induction.

Second Period Let us start from [Stage 8'](#). This stage is analogous to the baseline model. The consumer's consumption plan is as in (3.2). Equilibrium prices are as in [Appendix A.1](#). Let us now consider the lobbying equilibrium ([Stage 6'](#), [Stage 7'](#)). Suppose that a Good P is in office. Then, they always implement the policy. Suppose

⁴⁵In the baseline model, we assume that V is already expressed in present-value terms.

that a Bad P is in office. Their optimal contract is:

$$a'^* (l'_1 = l'_2 = 1, t'^* = \Delta\pi(m)) = 0 . \quad (\text{C.9})$$

A Bad P commits not to implement the policy $a' = 0$ if and only if both firms lobby. It is optimal for them to set the price of lobbying so as to extract all the surplus from the two firms: $t'^* = \Delta\pi(m)$. Since the game ends at the end of the period, a Bad P has no re-election incentives, and they never implement the pro-competitive reform. Firms are indifferent between lobbying or not, and, as in the baseline model, we assume that the tie is broken in favor of lobbying.⁴⁶

First Period Let us start from the voting stage (Stage 5'). If the consumer appoints a Bad P, the reform is never implemented, and they also bear a penalty $-\phi'$. If the consumer appoints a Good P, the reform is always implemented. Hence, the re-election rule described in Section 3.1 is optimal. The consumer re-elects the incumbent P if and only if $\hat{q} \geq q$, where $\hat{q}$ is the posterior belief that $\theta_P = \text{Good}$.

In Stage 4', the equilibrium is analogous to the second period, and to the baseline model.⁴⁷ We now turn our attention to the first-period lobbying equilibrium (Stage 2'-Stage 3'). A Good P always implements the pro-competitive reform. A Bad P faces a trade-off between accepting bribes today or ensuring re-election and bribes tomorrow. The solution of this trade-off depends on the discount factor δ .

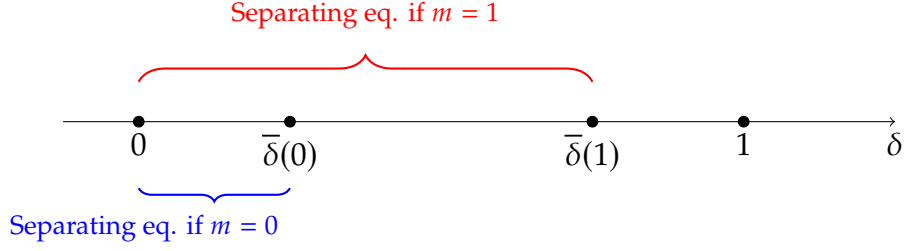
The optimal contract is analogous to (C.9). This contract is IC for the two firms. In particular, as $\delta < 1$, firms have no incentives to give up protection today ($a = 0$) to induce the re-election of the Bad P and gain protection tomorrow ($a' = 0$). However, for this to be IR for P, it must be that:

$$2\Delta\pi(m) \geq \delta (2\Delta\pi(m) + V') . \quad (\text{C.10})$$

⁴⁶If that was not the case, the optimal contract for P would be such that $t'^* = \Delta\pi(m) - \epsilon$, with $\epsilon \rightarrow 0_+$.

⁴⁷Firms and the consumer have no incentives to change their market behavior to induce changes in the election stage. The two firms have no price strategy that may change the consumer/voter's beliefs. The consumer's payoff is additive-separable across the two periods, so their maximization problem can be seen as two separate ones.

Figure C.4: Existence of Separating Equilibrium with endogenous t, ϕ, V .



Accepting bribes today must be better than winning the office and getting bribes tomorrow. Unsurprisingly, this is optimal for P if and only if they are impatient enough. The IR constraint (C.10) implies:

$$\delta \leq \frac{2\Delta\pi(m)}{2\Delta\pi(m) + V'} := \bar{\delta}(m). \quad (\text{C.11})$$

As in the baseline model, if (C.10) is not satisfied, the equilibrium is Pooling: both Ps types implement the policy and firms do not lobby. If instead (C.10) holds, the equilibrium is Separating. This more general model produces a result equivalent to Proposition 1. The merger increases the likelihood that lobbying emerges in equilibrium as $\bar{\delta}(1) > \bar{\delta}(0)$. If $m = 1$, firms have more incentives to lobby, and a Bad P can ask for higher bribes. Since bribes today are more valuable than bribes tomorrow, the merger increases lobbying in the first period. Figure C.4 illustrates this finding.

Merger Assessment We now briefly discuss the welfare analysis of the merger. In particular, we show that if there are no benefits from separation other than those coming from the market ($\phi' = 0$), the optimal merger policy from the consumer's perspective is always stricter than the consumer welfare standard ($\mu^* \leq \bar{\mu}(0)$).

To see this, suppose by contradiction that $\mu^* > \bar{\mu}(0)$. Given the intuition presented in the main text, this is possible if and only if the benefits from separation are high enough. In this two-period model, we can micro-found these benefits. Benefits from separation come from the fact that, in the second period, the anti-competitive policy

$(a' = 0)$ is approved with a low probability: $(1 - q)^2$ in a Separating equilibrium and $1 - q$ in a Pooling one. However, this comes at the cost of having the same policy with probability $1 - q$ in the first period (versus a zero probability in the Pooling equilibrium). Since the value of today is always higher than the value of tomorrow ($\delta < 1$), then this cannot be optimal. On the contrary, if $\phi' > 0$, it can be the case that $\mu^* \geq \bar{\mu}(0)$ when ϕ' is high enough. The intuition is analogous to the baseline model.

Remarks There are three main takeaways from this extension. First, a two-period model with endogenous ϕ , V , and t displays the same intuitions of the simple baseline model presented in Section 2: the merger increases lobbying. Second, when P is allowed to specify t as part of the mechanism, the optimal contract is such that all the surplus from trade is extracted from the two firms. P chooses the highest possible t so that firms are indifferent between lobbying or not. Third, the optimal merger policy may be less strict than the consumer welfare standard if and only if there are benefits from screening other than those coming from the narrowly defined market, for example, if corruptible politicians can implement regulations detrimental to consumers/voters in other parts of the economy (out-of-market externalities).